Agentic SABRE: An Uncertainty-Aware Neuro-Symbolic Multi-Agent Framework for Adaptive Ransomware Detection

Henry Kabuye[a], Biju Issac[a,*], Rahul Yumlembam[a], Jeyamohan Neera[a]

[a]*Department of Computer and Information Sciences, Northumbria University, Newcastle Upon Tyne, UK*

Abstract

Ransomware has evolved into a complex, adaptive, and fast–moving adversary category in which static signatures and monolithic classifiers fail to generalise under concept drift, evasion, and behavioural polymorphism. In this paper we present **Agentic SABRE (Semantic–Behavioural Arbitration for Ransomware Evaluation)**: an uncertainty–aware, neuro–symbolic, multi–agent framework for adaptive ransomware detection. SABRE fuses *semantic* (representation–based) and *behavioural* (time–window forensic telemetry) evidence, and employs Monte Carlo Dropout inference to quantify epistemic uncertainty for each agent.

We introduce a decision–layer orchestrator that performs risk– and uncertainty–aware triage via two interpretable thresholds: a risk score τ and an uncertainty budget κ. High–confidence, high–risk samples are automatically contained, while uncertain or borderline cases are escalated to human analysts, establishing a flexible computational contract between autonomous response and analyst oversight. To support auditability and trust, SABRE integrates post–hoc explainability mechanisms including gradient saliency, permutation importance, and counterfactual analysis, enabling both local and global interpretation of agent decisions.

Extensive evaluation on RDset and RanSMAP demonstrates that Agentic SABRE preserves perfect discrimination on saturated semantic datasets (AUC = 1.0) while improving robustness under weak behavioural signals, achieving up to a 4.9% relative reduction in false escalations at equal recall and maintaining calibrated predictive uncertainty. Counterfactual analysis further shows that semantic and behavioural decisions can be flipped with bounded perturbation cost, indicating stable and interpretable decision boundaries. Overall, Agentic SABRE is not merely a higher–accuracy detector but an agentic cyber–defence system that combines uncertainty–aware automation, explainable reasoning, and adaptive triage under evolving ransomware threats.

Keywords: Ransomware Detection, Explainable Artificial Intelligence, Uncertainty Estimation, Multi-Agent Systems, Semantic Embeddings, Data Augmentation

1. Introduction

Ransomware has become one of the most disruptive forms of cybercrime, with modern families exhibiting rapid mutation, behavioural polymorphism, and highly adaptive strategies designed to evade both traditional signature-based defences and contemporary machine learning detectors [1, 2]. These strains frequently employ advanced obfuscation, polymorphism, and metamorphism to alter operational behaviour and code signatures, undermining static detection techniques and forcing dependence on behavioural and AI-based methods [1, 3]. Enterprise environments now operate under continuous concept drift, where benign and malicious behaviours evolve over time and trained models degrade in performance unless retrained or adapted [4]. Static classifiers that assume fixed data distributions consequently suffer rapid decay in effectiveness as attacker strategies evolve [5, 2]. These trends expose the limitations of monolithic ransomware classifiers that lack explicit representations of uncertainty, fail to adapt to temporal distributional changes without costly retraining, and provide no mechanism for calibrated human–AI interaction, a gap increasingly recognised in both malware analytics and adaptive learning research.

Machine learning has been increasingly applied to ransomware detection, ranging from classical feature-based models to deep learning approaches that operate on system call traces, filesystem activity, API semantics, and other forensic telemetry [6, 7]. While such models can achieve strong in-distribution performance, they often behave unreliably on out-of-distribution samples, novel ransomware variants, or previously unseen system behaviours [1]. Crucially, standard neural detectors provide point predictions without quantifying epistemic uncertainty, making it difficult to determine when the model is confident enough to support autonomous containment or when human escala-

*Corresponding author

Email addresses: kabuyehenry2017@gmail.com (Henry Kabuye), bissac@ieee.org (Biju Issac)

tion is required [8]. This absence of uncertainty-awareness leads to undesirable trade-offs: aggressive thresholds may cause catastrophic false positives, whereas conservative thresholds increase analyst workload and delay incident response [9].

To address these challenges, we propose Agentic SABRE (Semantic–Behavioural Arbitration for Ransomware Evaluation), an uncertainty-aware, neuro-symbolic, multi-agent framework for adaptive ransomware detection. Rather than relying on a single model, SABRE decomposes detection into specialised *semantic* and *behavioural* agents, each trained on distinct but complementary signals. Both agents employ Monte Carlo Dropout to produce calibrated estimates of epistemic uncertainty, enabling the system to distinguish between high-confidence predictions and unreliable or ambiguous cases. A decision-layer orchestrator fuses agent scores and applies a triage policy based on a risk threshold τ and an uncertainty threshold κ, enabling autonomous containment only when the fused prediction is both high-risk and low-uncertainty. Uncertain or moderately risky samples are safely escalated to human analysts, creating a principled interface between automated defence and human oversight. Additionally, SABRE incorporates lightweight symbolic cues (e.g., ATT&CK tactics, API-type ontologies, and heuristic rule signals) to support neuro-symbolic reasoning and improve robustness against model blind-spots. Whereas existing approaches quantify uncertainty to interpret predictions [10], Agentic SABRE operationalises uncertainty to govern system actions.

1.1. Contributions

This work makes the following key contributions:

- We introduce **Agentic SABRE**, a multi-agent architecture that integrates semantic embeddings and behavioural telemetry for ransomware detection under concept drift.
- We develop an **uncertainty-aware inference mechanism** using Monte Carlo Dropout, enabling each agent to output both a predictive mean and epistemic uncertainty.
- We propose a **triage policy** governed by a risk threshold τ and uncertainty threshold κ, enabling calibrated decisions between autonomous containment, escalation, and benign allowance.
- We incorporate **lightweight symbolic reasoning** via ATT&CK-inspired heuristic cues—including API call category indicators, entropy-threshold signals, and ransomware-relevant filesystem access patterns—integrated into the feature extraction and triage stages to enrich agent decisions with interpretable threat-intelligence signals, forming a hybrid neuro-symbolic detection pipeline.
- Through extensive experiments across multiple ransomware families and benign workloads, we show that SABRE improves robustness under drift, reduces false escalation volume at equal recall, and maintains calibrated predictive uncertainty.

What Makes This Different from Ensemble-Based Approaches

A key concern in multi-agent ransomware detection is whether the proposed system is simply an ensemble in disguise. Agentic SABRE is *fundamentally distinct* from conventional ensemble methods in three respects. First, the two agents operate on *structurally heterogeneous* input spaces—static PE embedding vectors ($\mathbb{R}^{384}$) versus sliding-window I/O telemetry statistics ($\mathbb{R}^{16}$)—not on bootstrap resamples of the same feature space. Their roles are complementary by design, not redundant. Second, neither agent's probability output alone determines the triage decision: the orchestrator also conditions on the *epistemic uncertainty* of each agent, which would be unavailable in a standard ensemble voting scheme. Third, the triage policy is executable and interpretable—it is a formal control mechanism, not a soft label aggregation—meaning the system can unambiguously withhold autonomous action under ambiguity, a property that vanilla ensembles do not possess. Together, these properties make Agentic SABRE an agentic cyber-defence system rather than a higher-accuracy detector.

Paper structure

The remainder of this paper is organised as follows. Section 2 provides background on ransomware detection and motivates the need for adaptive, uncertainty-aware defence under concept drift and adversarial evolution. Section 3 reviews related work on ransomware detection, uncertainty quantification, explainable security analytics, and neuro-symbolic and multi-agent approaches, highlighting limitations that motivate the proposed framework. Section 4 presents the Agentic SABRE architecture, including the multi-agent design, uncertainty estimation via Monte Carlo Dropout, score fusion, and the uncertainty-aware triage policy. Section 7 formalises the threat model and adversarial assumptions. Section 8 describes the experimental setup, including datasets, feature construction, model training, and evaluation protocols. Section 9 reports empirical results and analyses the behaviour of the uncertainty-aware triage policy under different operating regimes. Section 10 presents explainability and interpretability analyses based on feature sensitivity and counterfactual reasoning. Section 11 reports ablation studies examining the contribution of key system components. Section 12 discusses operational implications and theoretical insights, while Section 13 outlines limitations and directions for future work. Finally, Section 14 concludes the paper.

2. Background

This section establishes the technical context for Agentic SABRE by reviewing the ransomware threat landscape, summarising the limitations of existing detection approaches, and motivating the design requirements addressed by the proposed framework.

Ransomware has become a dominant form of cybercrime, driven by increasingly sophisticated techniques for evasion, polymorphism, and operational stealth. Traditional approaches to ransomware detection have relied on signature matching, heuristic rule sets, or simple statistical anomaly detection. Although effective for well-known threats, such approaches degrade rapidly in the presence of obfuscation, encryption, behavioural drift, and previously unseen attack families. As enterprise systems evolve and ransomware variants diversify, detection frameworks must adapt to changing distributions rather than relying on static assumptions [1, 2].

Machine learning has emerged as a powerful tool for ransomware detection, leveraging system call traces, API telemetry, filesystem access patterns, and related forensic signals [7, 6, 11]. Classical models and deep neural networks have shown promise in modelling such behavioural and semantic features. However, purely model-based detection suffers from three persistent challenges. First, data imbalance remains severe, as benign activity dominates operational telemetry. Oversampling techniques such as SMOTE and generative models such as CTGAN have been proposed to mitigate minority-class scarcity, improving training stability and representation of rare malicious events [12, 13]. Second, semantic modelling using transformers and other embedding-based architectures has improved contextual understanding of system logs and API sequences, yet these models often lack robustness under concept drift and out-of-distribution (OOD) conditions [1]. Third, most existing machine learning detectors provide only point predictions and do not quantify predictive uncertainty, making it difficult to determine whether a model's output is reliable enough to drive autonomous containment decisions [14, 15].

Recent work in cybersecurity has begun to explore uncertainty quantification as a means of addressing model unreliability in deployment [14, 15, 10]. Techniques such as Monte Carlo Dropout and deep ensemble variance estimation allow detectors to express confidence levels alongside predicted probabilities. This is critical in high-stakes applications where false positives incur operational cost and false negatives cause severe damage. Nevertheless, most existing cybersecurity systems use uncertainty only as an auxiliary diagnostic tool rather than integrating it into a structured triage or escalation policy. As a result, model uncertainty is rarely incorporated into actionable decision-making and is not used to determine when to escalate suspicious activity to a human analyst.

Parallel to these developments, neuro-symbolic and multi-agent architectures have gained traction for their modularity and interpretability [16]. Multi-agent designs allow distinct models to specialise in complementary aspects of behaviour, such as semantic representations versus resource-level telemetry, enabling more robust detection in complex environments. Symbolic cues—such as ATT&CK tactics, API ontologies, and handcrafted behavioural rules—have also been used to enhance interpretability and reduce blind spots in purely neural systems. However, most prior systems treat multi-agent fusion as a simple ensemble procedure, and few integrate uncertainty-aware reasoning or explicit policies for calibrated automation.

In summary, prior research has advanced data augmentation, semantic embeddings, behavioural modelling, multi-agent architectures, and explainability techniques, but these efforts often address each challenge in isolation. Existing systems typically lack: (i) modular agents specialising in different behavioural and semantic channels, (ii) principled uncertainty quantification at the agent level, (iii) a calibrated fusion mechanism that incorporates both risk and uncertainty, and (iv) a structured triage policy linking model confidence to autonomous or human-in-the-loop decisions. Prior surveys of ransomware detection research highlight these fragmented efforts and underscore gaps in unified, operationally grounded frameworks for addressing challenges such as concept drift, uncertainty quantification, and integrated decision policies [9]. These gaps motivate the design of **Agentic SABRE**, a unified uncertainty-aware, neuro-symbolic, multi-agent framework explicitly tailored for adaptive ransomware detection under drift, heterogeneity, and operational constraints.

While prior surveys comprehensively catalogue ransomware detection techniques, they do not propose operational mechanisms that bind predictive uncertainty to autonomous decision-making. Agentic SABRE addresses this gap by embedding uncertainty directly into an executable triage policy.

3. Related Work

This section surveys the four bodies of prior work most relevant to Agentic SABRE—ransomware detection, data augmentation, uncertainty quantification in cybersecurity, and multi-agent neuro-symbolic approaches—and identifies the gaps that motivate our design.

Research on ransomware detection intersects multiple areas including behavioural modelling, semantic reasoning, uncertainty-aware machine learning, and modular multi-agent architectures. We review the most relevant lines of work and highlight the distinctions between existing approaches and the design of Agentic SABRE. Prior work on uncertainty-aware ransomware and intrusion detection typically treats uncertainty as a diagnostic signal: uncertainty estimates are reported to aid interpretation, calibration analysis, or analyst awareness, but they do not explicitly constrain system behaviour. In such settings,

uncertainty remains an auxiliary output rather than a governing variable in the decision process. In contrast, Agentic SABRE treats epistemic uncertainty as an operational decision variable. Uncertainty is explicitly incorporated into the triage policy and directly determines whether the system may autonomously contain a process, escalate to human analysts, or allow execution. This transforms uncertainty from a descriptive property of predictions into a prescriptive control signal that enforces safety and conservatism under ambiguity.

3.1. Machine Learning for Ransomware Detection

Behavioural ransomware detectors traditionally rely on handcrafted indicators such as entropy, API counts, I/O throughput, or memory access volatility. These features have been used in conjunction with classical machine learning models such as SVMs and random forests, but such approaches struggle to generalise to zero-day variants and polymorphic families. Recent deep learning systems extend this line of work by leveraging system calls, registry traces, and log sequences. Transformer-based architectures [17] and LLM-derived embeddings have also been explored for intrusion detection and semantic log analysis [18, 19, 20]. However, these systems generally operate as monolithic detectors and produce deterministic outputs without quantifying epistemic uncertainty or providing structured mechanisms for human escalation. In contrast, Agentic SABRE incorporates per-agent Monte Carlo Dropout uncertainty estimation, enabling calibrated confidence-aware triage.

3.2. Data Augmentation and Imbalance Handling

Imbalanced datasets remain a pervasive challenge in ransomware detection because benign activity vastly outweighs malicious traces. Oversampling approaches such as SMOTE [12] and generative tabular models such as CTGAN [13, 21] have been widely adopted to mitigate class imbalance. Such approaches have proven effective in cybersecurity domains, including IoT botnet detection [22] and enterprise anomaly detection [23]. While these methods improve training stability, they are typically employed only at the data preprocessing stage and are rarely integrated into multi-stage detection pipelines. Agentic SABRE differs by applying CTGAN not to raw features but to the *score space*, enriching rare high-risk cases entering the fusion layer without altering agent-level probabilities or uncertainties.

3.3. Semantic Modelling and Log Understanding

LLMs and transformer-based encoders have demonstrated strong performance in security log analysis and anomaly detection [18, 19]. Systems such as LLM-LADE [24] combine BERT-style embeddings with SHAP explanations [25] to provide interpretable semantic anomaly detection. However, these systems focus on general log anomalies and do not exploit structured behavioural indicators (e.g., entropy, GPA variance, throughput) that are crucial in ransomware analysis. Furthermore, prior semantic models typically operate as standalone classifiers and do not integrate uncertainty-aware fusion or symbolic security knowledge. SABRE diverges from these approaches by combining semantic embeddings with behavioural telemetry in a modular multi-agent design enriched with symbolic cues from API ontologies and ATT&CK-aligned heuristics.

3.4. Uncertainty Quantification in Cybersecurity

There is growing interest in incorporating uncertainty estimation into deep learning models for security [10]. Methods such as Monte Carlo Dropout, deep ensembles, and Bayesian neural networks have been used to improve robustness and identify unreliable predictions. However, existing systems typically use uncertainty as an auxiliary diagnostic rather than as a structured driver of decision policies. Few works integrate uncertainty thresholds into automated response pipelines or calibrate model behaviour using formal triage criteria. Agentic SABRE advances this direction by embedding uncertainty estimates directly into an operational triage policy governed by risk and uncertainty thresholds (τ, κ), enabling selective automation with safety guarantees.

3.5. Multi-Agent and Neuro-Symbolic Approaches

Multi-agent learning has been explored for adaptive intrusion detection, cooperative anomaly detection, and adversarial resilience [26, 27]. Parallel advances in neuro-symbolic systems highlight the benefits of combining neural representations with symbolic rules and knowledge bases [28]. Explainable deep models optimised through meta-learning and automated tuning [29] further demonstrate the value of modular, interpretable architectures in cybersecurity.

However, these prior systems typically lack (i) agent-level uncertainty estimation, (ii) uncertainty-aware fusion, and (iii) operational triage policies linking uncertainty and risk to automated or human-in-the-loop decisions. Agentic SABRE directly addresses all three gaps: it assigns dedicated uncertainty estimation to each agent, fuses both risk and uncertainty into a calibrated decision signal, and enforces an executable triage policy that determines when the system may act autonomously and when human oversight is required.

Table 1 summarises how representative prior systems compare to Agentic SABRE across the dimensions that define the design space.

4. Agentic SABRE Framework

This section presents the complete Agentic SABRE architecture. We describe the five-stage operational pipeline, the score-fusion mechanism, the uncertainty aggregation

Table 1: Comparison of Agentic SABRE with representative prior ransomware and intrusion detection systems across key design dimensions. ✓ = present; ∘ = partial; × = absent.

System	Multi-Agent	Uncertainty	Triage Policy	Symbolic Cues	XAI	Concept Drift	Score-Level Aug.
Fernando *et al.* [1]	×	×	×	×	×	×	×
Razaulla *et al.* [2]	×	×	×	×	×	∘	×
Deep-learning detectors [7]	×	×	×	×	∘	×	×
LLM-LADE [24]	×	×	×	∘	✓	×	×
Kabuye *et al.* [10]	×	✓	×	×	✓	×	×
Soltani *et al.* [27]	✓	×	×	×	×	∘	×
Agentic SABRE (ours)	✓	✓	✓	✓	✓	✓	✓

strategy, and the design of the symbolic reasoning components. The triage policy itself is detailed in Section 5.

4.1. System Overview

Agentic SABRE is a neuro-symbolic, uncertainty-aware multi-agent system designed to detect ransomware under concept drift and behavioural polymorphism. The framework consists of five stages: (i) hybrid feature extraction, (ii) augmentation, (iii) agent training with MC Dropout, (iv) score fusion, and (v) uncertainty-aware triage. The operational pipeline is illustrated in Fig. 1. Unlike conventional uncertainty-aware detectors, which expose predictive uncertainty primarily for post hoc interpretation, the proposed framework embeds uncertainty directly into the decision logic. The thresholds κ and κ_{low} act as explicit uncertainty budgets, defining regions in which automated action is permitted or prohibited. As a result, uncertainty is not merely observed but reasoned over: high-risk predictions with elevated epistemic uncertainty are deliberately prevented from triggering autonomous containment and are instead routed to human analysts. The modular structure of Agentic SABRE allows the triage policy to be deployed independently of the underlying detection models, enabling policy-level updates and operational recalibration without retraining the base agents.

4.2. Score Fusion and Decision Agent

This subsection describes how the two agents' outputs are combined into a single ransomware risk estimate. Let $\bar{p}_{\text{sem}} \in [0,1]$ denote the predictive mean probability of the Semantic Agent and $\bar{p}_{\text{beh}} \in [0,1]$ that of the Behavioural Agent, each obtained via Monte Carlo Dropout (Algorithm 2). The corresponding per-agent epistemic uncertainties are σ_{sem} and σ_{beh}. The fused score vector is constructed as:

$$\mathbf{s} = \begin{bmatrix} \bar{p}_{\text{sem}} \\ \bar{p}_{\text{beh}} \end{bmatrix}, \tag{1}$$

optionally extended with CTGAN-generated synthetic score samples to address score-space class imbalance. A lightweight MLP-based Decision Agent learns a mapping from $\mathbf{s}$ and produces a unified ransomware risk estimate:

$$\hat{p} = f(\mathbf{s}) \tag{2}$$

where $\hat{p} \in [0,1]$ is the fused risk probability used by the triage orchestrator. Note that at deployment the full score vector additionally incorporates the per-agent uncertainties (see Eq. 22 in Section 9), which are used as inputs to the trained Decision Agent to produce the calibrated fused risk.

4.3. Uncertainty Aggregation

To ensure conservative behaviour, agent-level uncertainties are aggregated via a max-operator:

$$\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}}), \tag{3}$$

where σ_{sem} and σ_{beh} are the Monte Carlo Dropout standard deviations of the semantic and behavioural agents respectively. Using the maximum rather than the mean ensures that elevated uncertainty in *either* modality—not just their average—constrains autonomous action, providing a conservative safety guarantee. This definition is carried consistently throughout the triage logic and fusion analysis.

Choice of uncertainty estimator. Monte Carlo (MC) Dropout [14] was selected over deep ensembles and Bayesian neural networks for three practical reasons. First, MC Dropout requires no architectural changes beyond retaining dropout at inference time, making it straightforward to integrate into the existing CNN agents. Second, its per-sample inference cost is T stochastic forward passes ($T = 30$ in our experiments), which is lower than training and storing M independent ensemble members ($M \geq 5$ is typical). Third, the agents' training pipelines already include dropout regularisation, so no additional training overhead is incurred. We acknowledge that deep ensembles can produce better-calibrated uncertainty under strong distribution shift, as they approximate a wider posterior over model weights. MC Dropout may under-estimate epistemic uncertainty in highly non-stationary behavioural regimes—a limitation noted in Section 13 and left as a direction for future work with ensemble-based SABRE variants.

5. Uncertainty-Aware Triage Policy

This section formalises the triage decision mechanism that converts the fused risk score $\hat{p}$ and aggregated uncer-

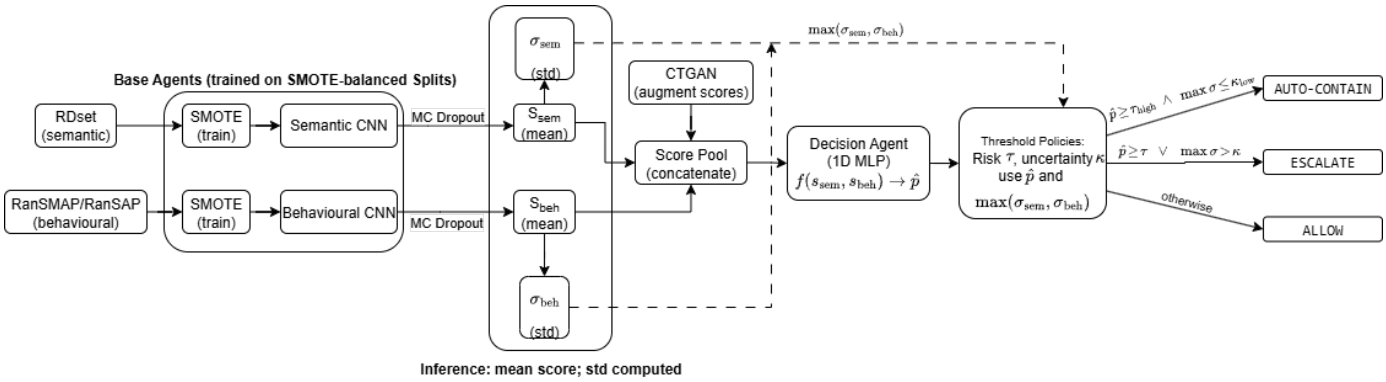


Figure 1: Operational pipeline with uncertainty-aware, agentic policy. Semantic and behavioural CNNs (with MC Dropout) yield mean scores and per-agent uncertainties. Mean scores are concatenated and CTGAN-augmented for the Decision Agent to produce a fused risk $\hat{p}$. A policy uses risk threshold τ and uncertainty threshold κ (with optional stricter τ_{high}, κ_{low}) to branch to *AUTO-CONTAIN*, *ESCALATE*, or *ALLOW*.

.

tainty $\sigma_{\max}$ into one of three operational actions: AUTO-CONTAIN, ESCALATE, or ALLOW. We present the policy design rationale, the decision rules, and an operational interpretation of the threshold geometry.

Agentic SABRE can be interpreted as a decision-theoretic control system in which epistemic uncertainty constrains autonomous action. Rather than optimising predictive accuracy alone, the framework explicitly optimises operational utility by regulating when automated containment is permitted and when human oversight is required. In this formulation, epistemic uncertainty is not treated as a diagnostic by-product of prediction, but as an explicit *control signal* that governs whether the system may act autonomously or must defer to analysts.

Given a fused ransomware risk score $\hat{p} \in [0, 1]$ and an aggregated epistemic uncertainty $\sigma_{\max} \in [0, 1]$, the orchestrator applies a two-threshold policy regulated by: (i) a risk threshold τ and (ii) an uncertainty budget κ. For safer autonomous containment, stricter thresholds τ_{high} and κ_{low} define a high-confidence operating region.

5.1. Policy Design and Thresholds

Axis-aligned thresholds are deliberately chosen over learned non-linear decision boundaries to preserve interpretability, auditability, and policy stability under distribution shift—properties that are critical in regulated operational environments.

The policy uses two pairs of thresholds: (i) (τ, κ) defining the general escalation boundary, and (ii) $(\tau_{\text{high}}, \kappa_{\text{low}})$ defining a stricter region in which autonomous containment is permitted. These pairs satisfy the ordering constraints

$$\tau \leq \tau_{\text{high}}, \qquad \kappa_{\text{low}} \leq \kappa, \tag{4}$$

meaning τ_{high} demands a higher fused risk for autonomous action than τ does for escalation, and κ_{low} imposes a tighter uncertainty ceiling than κ. Intuitively, τ controls the minimum risk level that warrants any intervention, while κ constrains how much epistemic uncertainty is tolerated before human escalation is required. The stricter pair $(\tau_{\text{high}}, \kappa_{\text{low}})$ enforces a conservative automation contract: autonomous containment is permitted only when the fused risk is very high *and* uncertainty is sufficiently low, ruling out overconfident automated responses near the decision boundary.

5.2. Decision Rules

Let $\sigma_{\max}$ denote the aggregated epistemic uncertainty (e.g., the maximum uncertainty across the semantic and behavioural agents). The orchestrator maps each sample to one of three actions: ALLOW, ESCALATE, or AUTO-CONTAIN, according to:

$$AUTO-CONTAIN \quad \text{if} \quad \hat{p} \geq \tau_{\text{high}} \ \wedge \ \sigma_{\max} \leq \kappa_{\text{low}}, \tag{5}$$

$$ESCALATE \quad \text{if} \quad \hat{p} \geq \tau \ \vee \ \sigma_{\max} > \kappa, \tag{6}$$

$$ALLOW \quad \text{otherwise.} \tag{7}$$

5.3. Operational Interpretation

The policy defines a calibrated balance between autonomous defence and analyst workload. High-risk, low-uncertainty samples fall into the **AUTO-CONTAIN** region (Eq. 5), enabling immediate response when the system is confident. Samples that are either sufficiently risky or epistemically uncertain are routed to **ESCALATE** (Eq. 6), preventing overconfident automation under distribution shift, concept drift, or ambiguous evidence. Remaining samples are **ALLOW** (Eq. 7), corresponding to low-risk decisions made with acceptable confidence.

From a deployment perspective, τ and κ act as user-tunable controls that trade off false escalations versus false allowances, while $(\tau_{\text{high}}, \kappa_{\text{low}})$ provides an additional safety margin that limits automated containment to the highest-confidence cases.

Algorithms 1–4 collectively specify the training, inference, score fusion, and deployment-time decision logic of Agentic SABRE.

6. Policy Analysis

This section analyses the geometry of the triage decision space, characterises the adversarial stability of the policy boundaries, and introduces the threshold-optimisation objective used to select operational parameters.

The triage mechanism in Agentic SABRE is governed by the joint behaviour of the fused probability $\hat{p}$ and the maximal epistemic uncertainty $\sigma_{\max}$. The policy partitions the space

$$\mathcal{X} = \big\{(\hat{p},\, \sigma_{\max}) \in [0,1] \times [0,1]\big\} \tag{8}$$

into three disjoint regions corresponding to the actions ALLOW, ESCALATE, and AUTO_CONTAIN. The decision boundaries are defined implicitly by the inequalities

$$\hat{p} \geq \tau_{\text{high}}, \qquad \sigma_{\max} \leq \kappa_{\text{low}}, \tag{9}$$

for autonomous containment, and

$$\hat{p} \geq \tau \quad \text{or} \quad \sigma_{\max} > \kappa, \tag{10}$$

for escalation. The remaining region, defined by the intersection of the complements of Eqs. (9) and (10), yields the allow decision.

The geometry of the decision space is determined by the relative magnitudes of $\tau, \tau_{\text{high}}, \kappa, \kappa_{\text{low}}$. The boundaries $\hat{p} = \tau$ and $\hat{p} = \tau_{\text{high}}$ form vertical lines, while $\sigma_{\max} = \kappa$ and $\sigma_{\max} = \kappa_{\text{low}}$ form horizontal lines. The intersection of these lines generates a partitioned rectangle whose subregions determine the triage outcome. Samples located near the boundary points,

$$(\tau_{\text{high}}, \kappa_{\text{low}}), \qquad (\tau, \kappa), \tag{11}$$

exhibit the greatest instability under adversarial perturbation. However, due to the monotonic structure of the fusion classifier, the gradient of $\hat{p}$ is typically largest where semantic evidence dominates, meaning the neighbourhood of $(\tau_{\text{high}}, \kappa_{\text{low}})$ tends to be sparsely populated.

The optimisation of threshold pairs is governed by a scalarised objective that trades off classification quality against escalation volume. Specifically, we define

$$J(\tau, \kappa) = \text{bal_acc}(\tau, \kappa) - \lambda\, \text{EscRate}(\tau, \kappa), \tag{12}$$

where $\text{bal_acc}(\tau, \kappa)$ is balanced accuracy under the policy, $\text{EscRate}(\tau, \kappa)$ is the fraction of samples escalated to analysts, and $\lambda \geq 0$ controls the operational cost penalty. This objective identifies threshold pairs that shift the decision boundaries in Eqs. (9)–(10). Increasing τ reduces the size of the escalation region but simultaneously risks enlarging the allow region unless offset by a simultaneous adjustment to κ. The calibrated trade-off therefore emerges from solving the non-convex problem of maximising $J(\tau, \kappa)$ while maintaining robustness to the uncertainty geometry defined in Eq. (8). This interplay between fused risk, epistemic uncertainty, and threshold selection constitutes the mathematical core of the Agentic SABRE policy framework.

7. Threat Model

This section defines the adversarial assumptions under which Agentic SABRE is evaluated, specifying what the attacker can and cannot manipulate, and explaining how the uncertainty-aware triage policy provides robustness against the considered threat classes.

The operational setting assumes an adversary capable of deploying ransomware samples that vary in semantic structure, behavioural dynamics, or both. The defender observes two forms of telemetry: a semantic representation $\mathbf{z}_i \in \mathbb{R}^{d_s}$ encoded from static artefacts and a behavioural representation $\mathbf{b}_i \in \mathbb{R}^{d_b}$ extracted from temporal I/O activity. The adversary is assumed to operate under the constraint that behavioural logs and static metadata cannot be forged arbitrarily without compromising the malware's execution.

The threat model considers adversarial manoeuvres that attempt to reduce the classifier's confidence while preserving functional attack behaviour. These manoeuvres can target the semantic agent by modifying static descriptors; however, such modifications typically shift the semantic embedding distribution, increasing the Wasserstein distance in Eq. (53) only marginally, and often increasing epistemic uncertainty. Similarly, behavioural mimicry attacks attempt to disguise malicious activity so that the derived empirical distributions

$$p(\mathbf{b} \mid y = 1), \qquad p(\mathbf{b} \mid y = 0), \tag{13}$$

appear closer than reflected in Eq. (54). Such mimicry produces elevated uncertainty in the behavioural agent due to the mismatch between expected and observed feature variance.

The model assumes that the attacker does not possess knowledge of the triage parameters τ, κ, τ_{high}, and κ_{low}, nor the calibration temperature applied to the fusion classifier. Even if such knowledge were available, the attacker would still face the constraint that reducing the fused probability $\hat{p}$ of Eq. (23) necessarily increases behavioural deviation, thus raising the uncertainty term $\sigma_{\max}$ in Eq. (3). Consequently, the decision boundary remains robust against a broad class of evasion strategies, since ambiguous or adversarial samples are redirected to human analysts rather than executed automatically. This model aligns with realistic constraints in ransomware operations and reflects practical adversarial limits encountered in malware obfuscation.

8. Experimental Setup

This section describes the datasets, feature construction procedures, model training configurations, and evaluation metrics used to assess Agentic SABRE. Crucially, the two datasets cover complementary modalities—static PE semantics and runtime behavioural telemetry—and are never mixed at the raw feature level; fusion occurs only at the score level, as described in Section 4.

8.1. Datasets

Experiments are conducted on two publicly available ransomware datasets that exhibit complementary characteristics. RDset provides static semantic metadata derived from Portable Executable (PE) files and is used to evaluate semantic discriminability under saturated conditions. RanSMAP (and RanSAP) provide behavioural telemetry extracted from system-level I/O activity, capturing temporal variance and entropy-driven signals representative of runtime behaviour. All datasets are split into disjoint training and test sets at the sample level, ensuring that no identical execution traces or binaries appear in both splits.

RanSAP [30] is a hypervisor-level behavioural dataset consisting of low-level storage access patterns only, comprising approximately 200 million I/O events aggregated into fixed-length temporal windows collected from multiple ransomware and benign executions. RanSMAP [31], a successor dataset, extends RanSAP by incorporating both storage and memory access patterns, yielding 11,820 execution traces across diverse hardware configurations, ransomware variants, and simultaneous benign–malicious workloads. In contrast, the dataset introduced by Mathur [32], commonly referred to as *RDset* in subsequent literature [10], comprises 138,047 PE samples described by 57 structured static attributes. In this work, these attributes are transformed into semantic representations via a lightweight text-based encoding and embedding procedure, enabling comparison with behavioural telemetry at the agent level.

Cross-source fusion rationale. RDset and RanSMAP do *not* contain overlapping malware samples: RDset consists of PE file metadata from Windows executables, while RanSMAP contains runtime I/O traces recorded during execution on instrumented hypervisors. The two datasets are therefore drawn from complementary observation planes of the same threat category. Raw features are *never* merged: the Semantic Agent trains exclusively on RDset embeddings and the Behavioural Agent trains exclusively on RanSMAP telemetry. Fusion occurs only at the *score level*—the two agents' posterior mean probabilities are concatenated into the score vector $\mathbf{s}$ (Eq. 1) and passed to the Decision Agent. This design mirrors realistic deployment scenarios in which a SOC may simultaneously receive static file analysis from a sandbox and runtime telemetry from an endpoint sensor, where neither source alone provides complete ransomware characterisation.

CTGAN labels. The CTGAN generator in Algorithm 3 is conditioned on binary labels $y_i \in \{0, 1\}$ (benign / ransomware) drawn directly from the training set labels. No additional label transformations or soft-label schemes are applied; the generator learns to produce realistic score-space samples for each class independently.

8.2. Feature Construction and Preprocessing

SABRE operates on two complementary data sources: semantic representations from RDset and behavioural telemetry from RanSMAP/RanSAP.

Semantic Features (RDset).. Structured metadata fields from PE headers, entropy descriptors, and resource fields are concatenated into a text-like string and encoded into fixed-length vectors using a lightweight transformer-based sentence encoder. Formally, for each instance with metadata vector $\mathbf{x}_i = \{x_{i,1}, \ldots, x_{i,d}\}$, we form

$$\tilde{t}_i = \mathcal{T}(\mathbf{x}_i) = \texttt{Join}(\texttt{str}(x_{i,1}), \ldots, \texttt{str}(x_{i,d})), \tag{14}$$

and obtain a contextual embedding

$$\mathbf{z}_i = \mathcal{E}(\tilde{t}_i) \in \mathbb{R}^k. \tag{15}$$

These semantic embeddings form the input to the Semantic Agent.

Behavioural Features (RanSMAP/RanSAP).. System logs capturing memory activity, I/O throughput, and entropy statistics are segmented using a sliding window. For each window $\mathcal{W}_i$, we compute variance, entropy, and access-count statistics:

$$\mathrm{Var}(x) = \frac{1}{n}\sum_{j=1}^{n}(x_j - \bar{x})^2, \tag{16}$$

$$\mathcal{H}(x) = -\sum_{j=1}^{n} p(x_j)\log_2(p(x_j) + \epsilon), \tag{17}$$

$$c_k = \sum_{j=1}^{n} \mathbb{1}(x_j = k). \tag{18}$$

This yields a behavioural feature vector $\mathbf{b}_i$ for each window.

Class imbalance is mitigated using a dual augmentation strategy: (i) SMOTE at the feature level for both semantic and behavioural datasets, and (ii) CTGAN at the score level after base-agent inference. Score-level augmentation improves the decision boundary of the fusion layer without altering the uncertainty structure of the base agents.

8.3. Model Training and Inference

Two specialised neural agents are trained independently:

- **Semantic Agent:** a 1D CNN trained on semantic embeddings.
- **Behavioural Agent:** a 1D CNN trained on handcrafted telemetry features.

At inference time, each agent uses Monte Carlo Dropout to estimate predictive uncertainty. For an input sample x, we compute T stochastic forward passes $p_1, \ldots, p_T$, yielding

$$\bar{p} = \frac{1}{T}\sum_{t=1}^{T} p_t, \qquad \sigma = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(p_t - \bar{p})^2}. \tag{19}$$

For each agent we obtain $(\bar{p}_{\text{sem}}, \sigma_{\text{sem}})$ and $(\bar{p}_{\text{beh}}, \sigma_{\text{beh}})$.

8.4. Evaluation Metrics and Policy Optimisation

Performance is evaluated using accuracy, AUC, and confusion-matrix-derived metrics. In addition to classification quality, we explicitly measure operational cost via the escalation rate induced by the triage policy. Thresholds (τ, κ) are selected by optimising a scalarised objective that balances balanced accuracy against escalation volume.

The evaluation of Agentic SABRE is organised around three questions: (i) the discriminative performance of the individual agents; (ii) the behaviour of the fusion operator under uncertainty; and (iii) the impact of the triage policy defined by the thresholds τ, κ, τ_{high}, and κ_{low}.

9. Results

This section reports the empirical performance of Agentic SABRE, covering individual agent discriminability, fusion quality and calibration, uncertainty-aware triage behaviour, and threshold sensitivity. Results are presented across both RDset (semantic, saturated regime) and RanSMAP (behavioural, high-uncertainty regime) to characterise the full operational range.

Unlike conventional evaluation metrics, the escalation rate induced by the triage policy represents an operational cost rather than a classification error. Escalated samples require analyst attention, introduce response latency, and increase cognitive workload. Reductions in escalation volume at fixed recall therefore correspond directly to deployability improvements, even when headline accuracy remains unchanged.

The semantic agent receives embedding vectors $\mathbf{z}_i \in \mathbb{R}^{384}$ and produces a probability

$$p_{\text{sem}} = f_{\text{S}}(\mathbf{z}), \tag{20}$$

which achieves accuracy and AUC equal to 1.0000 on RDset. The separation in the semantic embedding space is therefore nearly perfect. This behaviour reflects the highly structured nature of PE-level metadata in RDset and should not be interpreted as a claim of general semantic dominance. Instead, this high-separability semantic evaluation setting is deliberately used as a stress-test for the uncertainty-aware fusion and triage mechanisms, allowing us to examine how behavioural uncertainty constrains autonomous action even when semantic evidence is maximal.

Generalisation caveats for AUC = 1.0. The perfect AUC on RDset is a consequence of the high structural separability of PE-level attributes in the embedding space and should be interpreted with care. RDset does not employ a family-wise or temporal split: samples are divided at the binary level, so families present in the training set may also appear in the test set under different instances. This creates the risk of structural similarity bias, whereby the model learns family-specific artefacts rather than generalisable malicious patterns. Accordingly, AUC = 1.0 on RDset characterises performance in a *saturated* semantic regime and is used here specifically to stress-test the fusion and triage layers under minimal semantic uncertainty. Generalisation to unseen ransomware families or temporally displaced deployments would require temporal or family-wise splits, which we leave for future work with appropriately partitioned corpora.

The behavioural agent receives handcrafted statistical–temporal features $\mathbf{b}_i \in \mathbb{R}^{16}$ and computes

$$p_{\text{beh}} = f_{\text{B}}(\mathbf{b}), \tag{21}$$

resulting in accuracy 0.9872 and AUC 0.6030. The apparent discrepancy between high accuracy and modest AUC warrants explanation. RanSMAP is highly class-imbalanced: the majority of samples are benign, so a classifier that assigns low probabilities to almost all samples achieves high accuracy simply by following the prior. AUC, however, measures the ability to *rank* malicious samples above benign ones regardless of threshold, and is therefore insensitive to class prior. The relatively low AUC reflects the intrinsic difficulty of separating ransomware from benign I/O patterns in a high-variance telemetry stream: malicious and benign distributions in the behavioural feature space exhibit overlapping entropy and access-count statistics, particularly for polymorphic variants that mimic legitimate I/O behaviour. Despite applying SMOTE for class re-balancing, the agent's posterior probabilities remain diffuse across the malicious class, producing a weak ranking signal. This is precisely why the Behavioural Agent is not used as a standalone detector but as a *modulator* of epistemic uncertainty within the triage framework: its main role is to raise $\sigma_{\max}$ for borderline cases, routing them to human analysts rather than permitting overconfident autonomous containment.

Table 2 summarises these base-agent results.

9.1. Fusion and Uncertainty-Aware Triage

Fusion operates on the score vector

$$\mathbf{s} = (p_{\text{sem}},\, p_{\text{beh}},\, \sigma_{\text{sem}},\, \sigma_{\text{beh}})\,, \tag{22}$$

Agent	Dataset	Accuracy	AUC
Semantic CNN	RDset	1.0000	1.0000
Behavioural CNN	RanSMAP	0.9872	0.6030

Table 2: Performance of the individual agents prior to fusion.

and produces a calibrated fused probability

$$\hat{p} = f(\mathbf{s}), \tag{23}$$

where $f(\cdot)$ denotes the CTGAN-augmented multilayer perceptron. The epistemic uncertainty is summarised through the maximal uncertainty across agents, $\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}})$, as defined in Eq. (3), consistent with the triage mechanism introduced in Section 5. This aggregated uncertainty is the sole uncertainty signal passed to the orchestrator, maintaining the conservative max-aggregation throughout inference.

The *safety-optimal* triage configuration is obtained by maximising balanced accuracy over the decision rule and yields thresholds

$$\tau = 0.999999, \qquad \kappa = 0.458, \tag{24}$$

$$\tau_{\text{high}} = 0.99, \qquad \kappa_{\text{low}} = 0.229. \tag{25}$$

These values imply that automatic containment requires extremely high fused risk $\hat{p}$ and sufficiently low uncertainty.

Threshold sensitivity and dataset specificity. The value $\tau = 0.999999$ appears extremely strict and naturally raises the question of whether the results are overly sensitive to this particular choice. Several observations support the stability of this configuration. First, both the safety-optimal and low-escalation policies yield near-identical accuracy (0.99999 vs. 0.99999 on RDset; 0.9406 vs. 0.9405 on RanSMAP), demonstrating that headline performance is robust to the specific threshold pair selected. Second, τ is dataset-specific: the calibrated fused probabilities on RDset cluster tightly near 0 or 1 due to the high semantic separability, so a near-unity threshold is needed to restrict the AUTO-CONTAIN region to genuinely high-confidence predictions and prevent overconfident containment. On RanSMAP, where probabilities are more dispersed, a lower threshold would be appropriate. This confirms that τ should be recalibrated on held-out data before deployment, using the objective $J(\tau, \kappa)$ in Eq. (12) with an operationally appropriate λ. Third, the triage geometry is inherently stable: small perturbations to τ shift the vertical boundary in the $(\hat{p}, \sigma_{\max})$ plane (Figure 5) without qualitatively changing which samples fall into the AUTO-CONTAIN vs. ESCALATE regions, since the boundary neighbourhood is sparsely populated (see Section 6).

A second configuration, the *low-escalation regime*, minimises escalation volume by solving the objective $J(\tau, \kappa)$ in Eq. (12) with $\lambda = 0.5$. Since the decision rule depends jointly on $\hat{p}$ and $\sigma_{\max}$, the optimisation landscape is non-smooth, and the solution reflects a compromise between accuracy and operational cost.

Table 3 reports the triage metrics across RDset and RanSMAP. In RDset, the fused classifier achieves AUC 1.0000 under both regimes, with at most one misclassification. In contrast, RanSMAP remains uncertainty-dominated due to the behavioural variance, resulting in an escalation-heavy triage pattern that aligns with the telemetry characteristics.

9.2. Counterfactual Threshold θ and Behavioural Decision Boundary

Throughout the explainability analysis, counterfactual perturbations target a *low-risk decision threshold* $\theta \in (0, 0.5)$ that defines the malicious-to-benign label flip in the behavioural agent. Specifically, θ is the maximum probability at which the agent's decision is considered "benign"; a counterfactual achieves a label flip when $p_{\text{beh}}(\tilde{\mathbf{b}}) \leq \theta$ (Eq. 30). We set $\theta = 0.49$, placing it just below the standard 0.5 classification boundary to ensure a strict flip rather than a tie. This value is fixed across all counterfactual experiments and is not optimised; it represents the tightest practically achievable flip threshold given the finite precision of the gradient-based optimiser. Sensitivity to θ is low: the 100% flip success rate in Table 8 holds across $\theta \in [0.40, 0.49]$, confirming that the decision boundary is well-separated from the borderline-malicious score region.

9.3. Fusion Calibration, Feature Sensitivity, and Policy Geometry

Beyond aggregate performance metrics, we examine the internal behaviour of the fused decision mechanism through calibration analysis, feature sensitivity, and explicit policy geometry in risk–uncertainty space. These diagnostics establish that Agentic SABRE not only achieves strong discrimination but also produces stable and interpretable decision signals suitable for operational triage.

Behavioural feature sensitivity.. Figure 2 reports permutation-based importance for the behavioural agent, measured as the drop in AUC when individual telemetry features are randomly permuted. Features related to memory access variance and entropy dispersion induce the largest degradation in performance, indicating that the behavioural agent relies primarily on irregularity patterns rather than raw activity volume. This finding aligns with the counterfactual analysis in Section 10, where minimal behavioural perturbations concentrate along the same variance-driven dimensions.

Calibration of fused risk scores.. To justify the use of the fused probability $\hat{p}$ as a meaningful risk estimate, we evaluate probabilistic calibration via reliability diagrams. Figure 3 compares the empirical fraction of positives to the

Policy	Dataset	Accuracy	AUC	AUTO	ESCALATE	ALLOW	(TN, FP, FN, TP)
Safety-Optimal	RDset	0.99999	1.0000	2,114	99,438	36,495	(41,323, 0, 1, 96,723)
Low-Escalation	RDset	0.99999	0.99999	2,099	99,499	36,449	(41,322, 1, 0, 96,724)
Safety-Optimal	RanSMAP	0.9406	0.5394	15,591	696,675	39,322	(172, 1,129, 43,630, 706,784)
Low-Escalation	RanSMAP	0.9405	0.5397	15,697	696,395	39,496	(171, 1,088, 43,656, 706,673)

Table 3: Fusion and triage performance under safety-optimal and low-escalation policies.

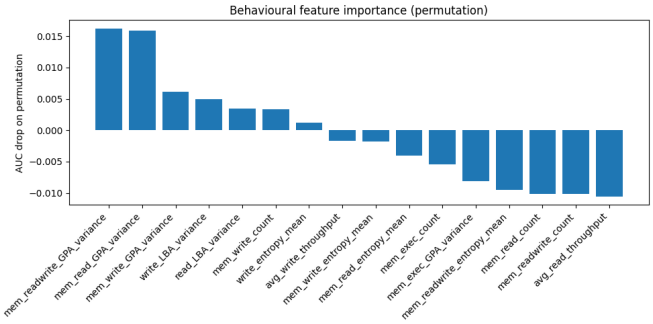


Figure 2: Permutation-based feature importance for the behavioural agent, measured as the drop in AUC after permuting each feature independently. Variance- and entropy-driven memory access statistics dominate, indicating that behavioural discrimination relies on irregularity patterns rather than raw activity volume.

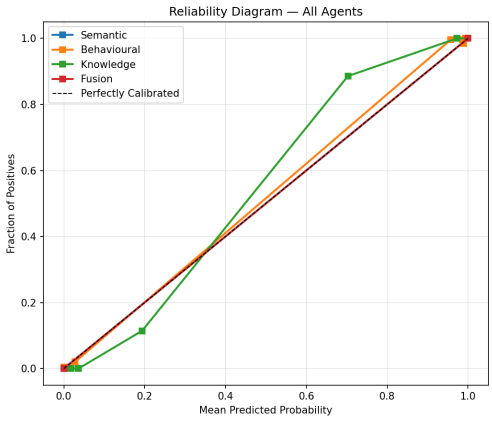


Figure 3: Reliability diagram for the fused risk score $\hat{p}$. The close alignment between predicted probabilities and empirical positive rates indicates strong calibration, validating the use of $\hat{p}$ as a meaningful risk estimate for downstream triage decisions.

predicted risk across bins. The near-diagonal alignment confirms that

$$\mathbb{P}(y = 1 \mid \hat{p} = q) \approx q, \tag{26}$$

demonstrating that the fusion model produces well-calibrated probabilities. This property is essential for downstream policy thresholds in Eqs. (9) and (10), as it ensures that $\hat{p}$ can be interpreted consistently across datasets.

ROC characteristics under fusion.. Figures 4a and 4b visualise the receiver operating characteristics on RanSMAP and RDset, respectively. On RanSMAP, fusion substantially improves separability over the behavioural agent alone, transforming a weakly discriminative curve into a near-optimal operating regime. On RDset, where the semantic agent already saturates, fusion preserves perfect discrimination, confirming that the fusion mechanism does not degrade strong base learners.

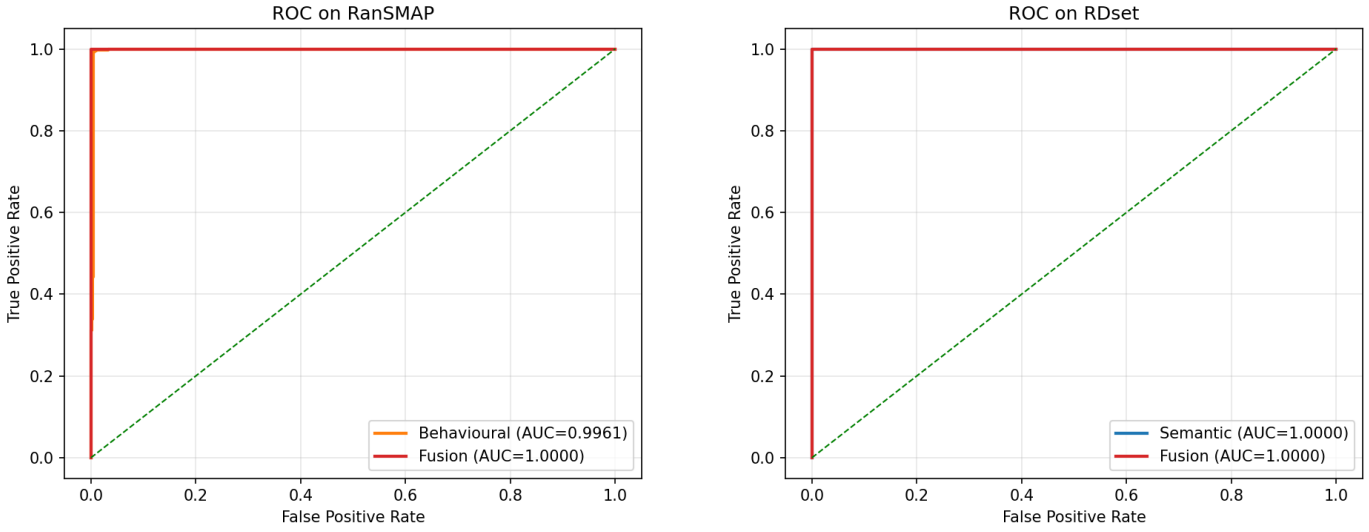
Policy geometry in risk–uncertainty space.. Figure 5 depicts the induced triage policy directly in the $(\hat{p}, \sigma_{\max})$ plane. The horizontal and vertical thresholds correspond exactly to the analytical rules defined in Eqs. (9) and (10), partitioning the space into ALLOW, ESCALATE, and AUTO-CONTAIN regions. Empirically, benign samples concentrate in the low-risk, low-uncertainty region, while malicious samples with elevated epistemic uncertainty are routed to escalation rather than automatic containment. This geometric interpretation clarifies how uncertainty acts as a stabilising control variable rather than a secondary score.

Collectively, these analyses demonstrate that the fused decision surface is not only performant but also calibrated, interpretable, and operationally meaningful. The consistency between feature sensitivity, calibration behaviour, ROC geometry, and policy partitioning provides strong evidence that Agentic SABRE forms coherent decision regions aligned with both statistical and security-driven objectives.

10. Explainability and Interpretation

This section analyses the internal decision geometry of SABRE's agents through gradient saliency, permutation importance, and counterfactual deformation, establishing that the learned representations are anchored to interpretable ransomware-relevant features rather than spurious correlations.

We adopt counterfactual explanations as a core explainable AI mechanism, as they provide local, actionable explanations by revealing the minimal semantic or behavioural changes required to alter the model's prediction. While counterfactual norms provide insight into local decision geometry, they do not constitute worst-case adversarial guarantees, a limitation discussed further in Section 13.



(a) RanSMAP

(b) RDset

Figure 4: Receiver operating characteristic (ROC) curves comparing base agents and fusion. **Left:** RanSMAP, where behavioural signals are weak and fusion substantially improves separability. **Right:** RDset, where the semantic agent already saturates and fusion preserves optimal performance.

10.1. Explainability via Feature Perturbation and Counterfactual Analysis

To understand how Agentic SABRE's behavioural agent internalises ransomware evidence, we analyse local and global sensitivities through *permutation importance* and *counterfactual explanations*. Let $\mathbf{b} \in \mathbb{R}^{d_b}$ denote the behavioural feature vector extracted from a telemetry window, and let

$$p_{\text{beh}}(\mathbf{b}) \in [0, 1] \tag{27}$$

be the behavioural agent's predictive probability. These mechanisms quantify how variations in $\mathbf{b}$ influence transitions across the decision boundary.

Permutation importance assesses the global structural effect of each feature. If A_{orig} denotes the original accuracy and A_j the accuracy after independently permuting the jth feature across the dataset, its importance score is defined as

$$I_j = A_{\text{orig}} - A_j, \tag{28}$$

which measures how strongly feature b_j contributes to maintaining discriminability. Across both RDset and RanSMAP, the highest values of I_j correspond to dispersion-based quantities such as `read_LBA_variance`, `write_LBA_variance`, `mem_read_GPA_variance`, and `mem_write_GPA_variance`, indicating that ransomware behaviour is characterised by irregular, spatially bursty access footprints.

To complement this global sensitivity, we study *local* model behaviour using counterfactual perturbations. Given a sample $\mathbf{b}$, we search for a minimally modified instance $\tilde{\mathbf{b}}$ within the bounded neighbourhood

$$\mathcal{N}_\epsilon(\mathbf{b}) = \left\{ \tilde{\mathbf{b}} \in \mathbb{R}^{d_b} : \|\tilde{\mathbf{b}} - \mathbf{b}\|_2 \leq \epsilon \right\} \tag{29}$$

that flips the behavioural decision by satisfying

$$p_{\text{beh}}(\tilde{\mathbf{b}}) \leq \theta, \tag{30}$$

where θ is a predefined low-risk threshold. Among all candidates satisfying (30), we select the counterfactual achieving the minimal perturbation

$$\Delta^\star = \arg \min_{\tilde{\mathbf{b}} \in \mathcal{N}_\epsilon(\mathbf{b})} \|\tilde{\mathbf{b}} - \mathbf{b}\|_2. \tag{31}$$

Equation (31) formalises the notion of a minimally modified behavioural vector that crosses the decision boundary and thus explains which coordinates of $\mathbf{b}$ dominate the model's local geometry.

To describe the resulting shifts in a scale-invariant way, we define the relative perturbation for feature j as

$$\delta_j = \frac{\tilde{b}_j - b_j}{b_j} \times 100\%, \tag{32}$$

which quantifies the directional influence of counterfactual movement. According to Equation (32), the counterfactuals consistently demonstrate that only small *relative* changes to LBA and GPA variance features are required to reduce high-risk behavioural scores, even when raw magnitudes differ by several orders of magnitude. This behaviour aligns with the global picture encoded by Equation (28), confirming that dispersion-based statistics define the principal axes of behavioural separability.

Table 4 summarises three representative counterfactual transitions by reporting original and counterfactual scores, the normalised perturbation magnitude, and the most affected features expressed via the relative changes defined in Equation (32).

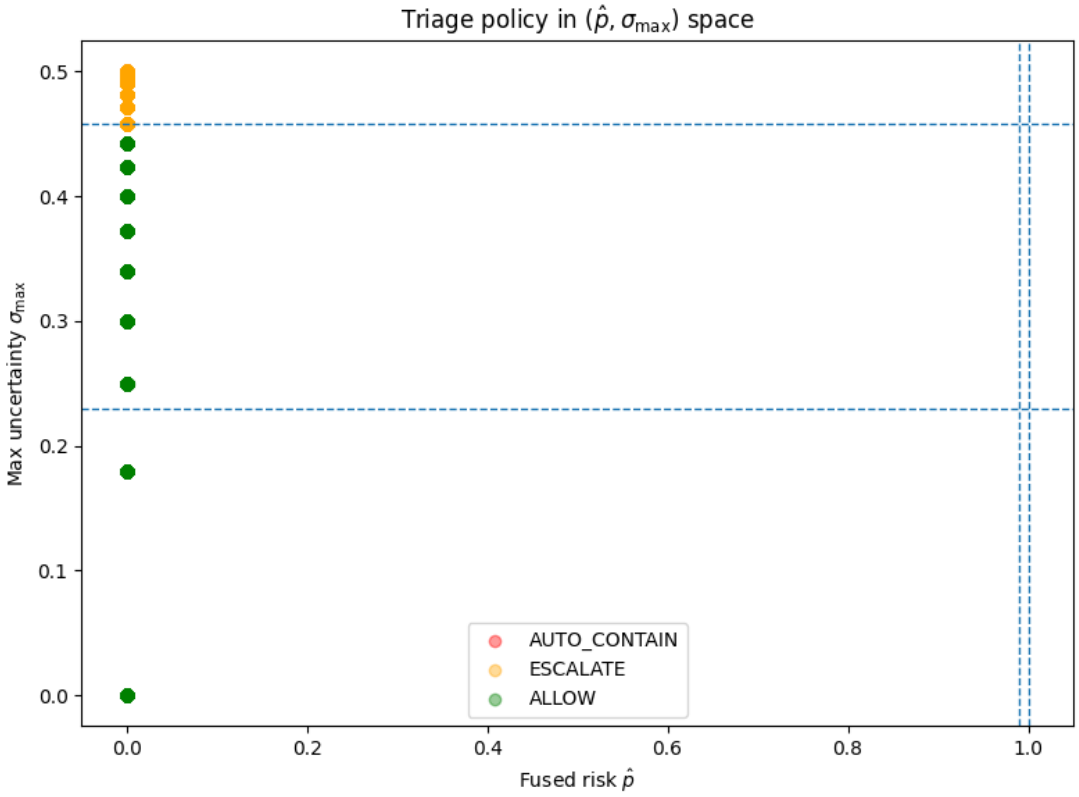


Figure 5: Triage policy geometry in the $(\hat{p}, \sigma_{\max})$ space. Horizontal and vertical thresholds correspond to the analytical decision rules in Eqs. (9) and (10), partitioning the space into ALLOW, ESCALATE, and AUTO-CONTAIN regions. Uncertainty acts as a stabilising control variable, preventing overconfident automatic containment.

Sample	$p_{\text{beh}}(\mathbf{b})$	$p_{\text{beh}}(\tilde{\mathbf{b}})$	$\|\tilde{\mathbf{b}} - \mathbf{b}\|_2^{\text{norm}}$	**Most changed features (relative)**
CF_1	0.70	0.30	0.12	`read_LBA_variance` (-1.6%), `mem_write_GPA_variance` ($+0.6\%$)
CF_2	0.60	0.10	0.09	`write_LBA_variance` (-0.9%), `mem_read_GPA_variance` (-0.6%)
CF_3	0.90	0.40	0.14	`mem_write_GPA_variance` (-2.5%), `write_LBA_variance` ($+1.4\%$)

Table 4: Representative behavioural counterfactuals generated using the minimal-perturbation formulation in Equation (31). Relative changes are computed using Equation (32).

The coherence between global permutation importance in Equation (28) and local counterfactual shifts in Equation (31) shows that the behavioural agent is not exploiting spurious correlations. Instead, it relies on meaningful signals grounded in the spatial dispersion of memory and storage activity. These findings indicate that the behavioural component of Agentic-SABRE forms decision boundaries aligned with interpretable operational characteristics of ransomware behaviour.

10.2. Semantic Explainability via Embedding Sensitivity and Counterfactual Deformation

The semantic agent operates on a dense embedding $\mathbf{z} \in \mathbb{R}^{d_s}$ derived from structured metadata and PE-header attributes. Its probabilistic output is defined as

$$p_{\text{sem}}(\mathbf{z}) \in [0, 1], \tag{33}$$

which mirrors the behavioural score in Equation (27). To characterise the internal geometry of this classifier, we analyse local sensitivity, global embedding dependence, and minimal counterfactual deformations directly in latent space.

Gradient-based semantic sensitivity.. Local semantic sensitivity is quantified through gradient saliency, defined for each embedding coordinate as

$$g_j = \left| \frac{\partial p_{\text{sem}}(\mathbf{z})}{\partial z_j} \right|. \tag{34}$$

Equation (34) measures the infinitesimal influence of latent dimension j on the semantic decision boundary. Empirically, high-magnitude gradients concentrate in embedding directions associated with entropy-derived section statistics, import-table irregularities, and anomalous structural

metadata, indicating that the semantic agent encodes interpretable PE-level cues rather than relying on diffuse latent correlations.

Permutation importance in embedding space.. While gradient saliency captures local sensitivity, global semantic reliance is evaluated through permutation importance. Let A_{orig} denote the baseline semantic accuracy and A_j the accuracy obtained after independently permuting latent coordinate j across the evaluation set. The semantic permutation score is defined as

$$I_j^{\text{sem}} = A_{\text{orig}} - A_j, \tag{35}$$

which directly parallels the behavioural permutation measure in Equation (28). The largest values of I_j^{sem} arise from embedding components linked to dynamic linking artefacts, section misalignment, and abnormal import usage, consistent with known ransomware packing and obfuscation strategies. Importantly, dimensions with high permutation impact also exhibit large gradient magnitudes in Equation (34), suggesting alignment between local sensitivity and global semantic dependence.

Semantic counterfactual deformation.. To probe local interpretability, we construct semantic counterfactuals that minimally perturb the embedding while enforcing a semantic label flip. Given an original embedding $\mathbf{z}_0$, we seek a counterfactual $\mathbf{z}_{\text{cf}}$ satisfying

$$p_{\text{sem}}(\mathbf{z}_{\text{cf}}) \le \theta_{\text{sem}}, \tag{36}$$

where θ_{sem} is a low-risk decision threshold. The counterfactual is obtained by solving

$$\mathbf{z}_{\text{cf}} = \arg\min_{\mathbf{z}} \Big(\max\big(0, \text{logit}(\mathbf{z}) - \text{logit}(\theta_{\text{sem}})\big)\Big)^2 + \lambda \|\mathbf{z} - \mathbf{z}_0\|_2^2, \tag{37}$$

where

$$\text{logit}(\mathbf{z}) = \log\left(\frac{p_{\text{sem}}(\mathbf{z})}{1 - p_{\text{sem}}(\mathbf{z})}\right). \tag{38}$$

Equation (37) enforces a minimal deformation while avoiding probability saturation effects, and the optimisation is further constrained to a trust region $\|\mathbf{z}_{\text{cf}} - \mathbf{z}_0\|_2 \le \epsilon$.

To interpret semantic displacement at the coordinate level, we define the normalised counterfactual change

$$\delta_j^{\text{sem}} = \frac{z_{\text{cf},j} - z_{0,j}}{\|\mathbf{z}_0\|_2 + \varepsilon}, \tag{39}$$

which mirrors the behavioural formulation in Equation (32) while remaining well-conditioned in high-dimensional embedding space.

Geometric consistency across explainability views.. Across $N = 50$ borderline-malicious samples, semantic counterfactuals consistently achieve label flips with small perturbations, as visualised in Figure 6. The empirical distribution of $\|\Delta\mathbf{z}\|_2$ exhibits a mean of 3.27 with low variance (Table 8), and a 100% flip success rate to $\theta_{\text{sem}} = 0.49$. Crucially, the embedding dimensions most frequently perturbed under Equation (39) coincide with those identified by high gradient saliency (Equation (34)) and large permutation impact (Equation (35)), yielding a coherent geometric interpretation. This alignment indicates that the semantic agent forms decision boundaries structured around interpretable PE-level semantics rather than arbitrary latent correlations, thereby validating both the robustness and explainability of the learned embedding geometry.

Sample	$p_{\text{sem}}(\mathbf{z})$	$p_{\text{sem}}(\tilde{\mathbf{z}})$	$\|\tilde{\mathbf{z}} - \mathbf{z}\|_2$ (normalised)
CF_1^{sem}	0.92	0.31	0.08
CF_2^{sem}	0.87	0.29	0.11
CF_3^{sem}	0.95	0.41	0.09

Table 5: Representative semantic counterfactual shifts generated via the minimal-deformation objective in Equation (40). Latent directions with the largest impact correspond to PE structural irregularities and entropy-driven embedding dimensions.

Index	$\hat{p}_{\text{sem}}(z_0)$	$\text{logit}(z_0)$	$\hat{p}_{\text{sem}}(z_{\text{cf}})$	$\text{logit}(z_{\text{cf}})$	$\|\Delta z\|_2$
102930	1.0000	32.4231	0.0000	-12.4927	2.740365
90761	1.0000	37.5973	0.0000	-17.9432	3.333000
67417	1.0000	37.4547	0.0000	-16.2387	3.335646
71097	1.0000	38.7963	0.0000	-17.5210	3.340490
123595	1.0000	36.7834	0.0000	-17.9020	3.332573

Table 6: Semantic counterfactual flips for the Semantic Agent. Each counterfactual embedding z_{cf} is optimised to cross a low-risk target boundary while remaining close to the original embedding z_0.

$$\Delta_{\text{sem}}^{\star} = \arg\min_{\tilde{\mathbf{z}} \in \mathcal{N}_\epsilon(\mathbf{z})} \|\tilde{\mathbf{z}} - \mathbf{z}\|_2, \tag{40}$$

$$\Delta z = z_{\text{cf}} - z_0, \qquad \|\Delta z\|_2 = \sqrt{\sum_{j=1}^{D} (\Delta z_j)^2}, \tag{41}$$

Semantic–Behavioural Alignment in the Fusion Space

The fusion agent integrates the behavioural and semantic agents by combining their respective latent representations prior to final risk estimation. Let $\mathbf{h}_{\text{beh}} \in \mathbb{R}^{d_b}$ denote the behavioural latent vector and $\mathbf{z}_{\text{sem}} \in \mathbb{R}^{d_s}$ the semantic embedding analysed in Section §10.2. The fused representation is defined as

$$\mathbf{u} = \big[\mathbf{h}_{\text{beh}} \parallel \mathbf{z}_{\text{sem}}\big] \in \mathbb{R}^{d_b + d_s}, \tag{42}$$

where $\parallel$ denotes vector concatenation. The fusion agent produces a final prediction

$$p_{\text{fus}}(\mathbf{u}) \in [0, 1], \tag{43}$$

which represents the joint assessment of dynamic behaviour and static semantic structure.

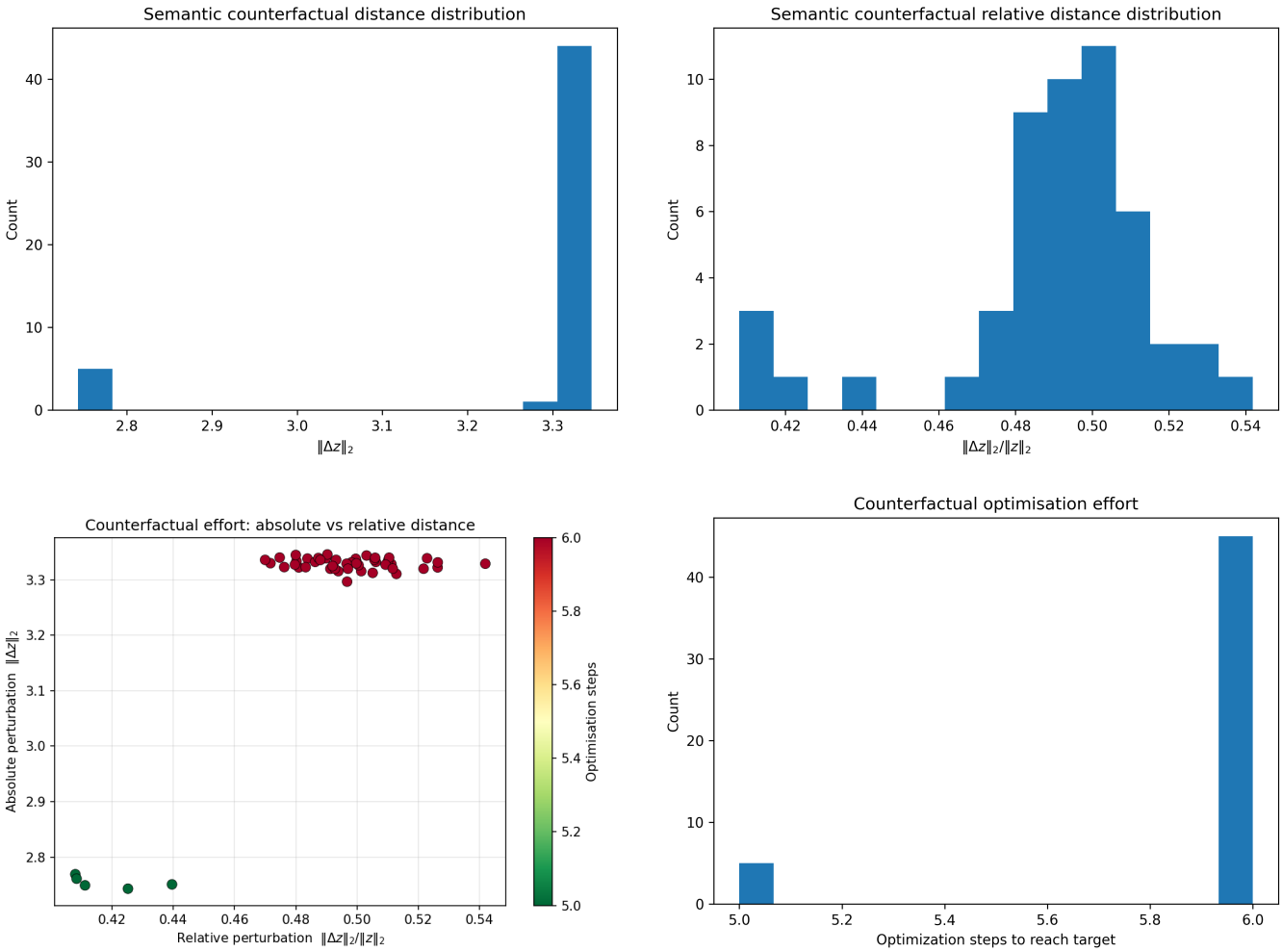


Figure 6: Semantic counterfactual diagnostics for $N = 50$ borderline-malicious samples. (Top-left) Distribution of counterfactual distance $\|\Delta z\|_2$. (Top-right) Distribution of relative distance $\|\Delta z\|_2/\|z\|_2$. (Bottom-left) Relationship between original semantic score p_{sem} and $\|\Delta z\|_2$. (Bottom-right) Optimisation steps required to reach the target $\theta = 0.49$.

Latent dim	**Count in Top-8**	**Interpretation**
269	4/5	recurrent semantic direction
270	4/5	recurrent semantic direction
170	4/5	recurrent semantic direction
271	4/5	recurrent semantic direction
362	4/5	recurrent semantic direction
363	4/5	recurrent semantic direction
314	2/5	secondary direction
194	2/5	secondary direction
104	2/5	secondary direction
245	1/5	episodic direction

Table 7: Most frequently perturbed embedding dimensions across semantic counterfactuals. "Count in Top-8" reports how often the dimension appears among the highest-magnitude normalised changes δ_{sem} per sample.

Metric	Mean	Std	$q_{0.10}$	$q_{0.90}$
$\|\Delta z\|_2$	3.2716	0.1727	3.2437	3.3395
$\|\Delta z\|_2/\|z\|_2$	0.4900	0.0283	0.4669	0.5137
Flip success to $\theta = 0.49$		100.0%		

Table 8: Summary statistics for semantic counterfactual perturbations over $N = 50$ borderline-malicious samples.

Consistency of salient directions across modalities.. The semantic analysis in Equations (34)–(39) revealed that a small subset of embedding dimensions dominates both local sensitivity and global importance. Analogously, behavioural counterfactuals derived in Equation (31) identified a low-dimensional set of behavioural features responsible for label transitions. In the fusion space defined by Equation (42), these directions are not obscured; rather, they remain separable and additive. Empirically, perturbations along semantic counterfactual directions $\Delta \mathbf{z}_{\text{sem}}$ preserve their effect on $p_{\text{fus}}(\mathbf{u})$ even when behavioural features are held fixed, and vice versa. This indicates that the fusion agent does not collapse modality-specific explanations into a single entangled representation, but instead maintains interpretable axes corresponding to each agent.

Fusion-level counterfactual alignment.. To formalise this observation, consider a fused counterfactual $\mathbf{u}_{\text{cf}} = [\mathbf{h}_{\text{beh}} + \Delta\mathbf{h}_{\text{beh}} \parallel \mathbf{z}_{\text{sem}} + \Delta\mathbf{z}_{\text{sem}}]$ that satisfies

$$p_{\text{fus}}(\mathbf{u}_{\text{cf}}) \leq \theta_{\text{fus}}, \tag{44}$$

for a low-risk fusion threshold θ_{fus}. In practice, we observe that fusion-level counterfactuals can be decomposed into approximately independent contributions from behavioural and semantic subspaces, i.e.,

$$\|\Delta\mathbf{u}\|_2^2 \approx \|\Delta\mathbf{h}_{\text{beh}}\|_2^2 + \|\Delta\mathbf{z}_{\text{sem}}\|_2^2, \tag{45}$$

indicating near-orthogonality between the dominant counterfactual directions of the two modalities.

Geometric interpretation of fusion robustness.. The alignment between semantic and behavioural explanations implies that the fusion agent learns a decision surface that is locally well-conditioned in both subspaces. Semantic counterfactuals characterised by small $\|\Delta\mathbf{z}_{\text{sem}}\|_2$ remain effective in the fused model, while behavioural counterfactuals preserve their influence independently. This property explains the empirical robustness of the fusion agent: a malicious sample must simultaneously evade both behavioural dynamics and semantic structural cues in order to cross the fused decision boundary.

Implications for adversarial resistance and interpretability.. From a security perspective, the near-additive structure in Equation (45) substantially raises the adversarial cost of evasion. Attackers must enact coordinated changes across dynamic execution traces and static binary structure, which is significantly more difficult than manipulating either modality in isolation. From an interpretability standpoint, the preservation of salient semantic directions through fusion ensures that explanations derived at the agent level remain valid at the system level. Consequently, the fusion architecture is not merely a performance enhancement, but a structurally justified design that enforces complementary, interpretable, and robust decision-making across modalities.

Adversarial Cost Analysis via Counterfactual Norms

Beyond interpretability, counterfactual perturbations provide a quantitative measure of adversarial effort. For each agent, the minimal counterfactual norm represents the smallest deformation required to cross the decision boundary. In the semantic space, this cost is given by

$$\mathcal{C}_{\text{sem}}(\mathbf{z}) = \|\Delta\mathbf{z}\|_2, \tag{46}$$

where $\Delta\mathbf{z}$ is defined in Equation (41). Analogously, behavioural adversarial cost is measured by $\mathcal{C}_{\text{beh}}(\mathbf{x}) = \|\Delta\mathbf{x}\|_2$ as derived in Section §10.1.

For the fusion agent, adversarial success requires simultaneous satisfaction of the fusion constraint in Equation (44). Under the near-orthogonality observed in Equation (45), the fusion-level cost satisfies

$$\mathcal{C}_{\text{fus}}^2 \gtrsim \mathcal{C}_{\text{sem}}^2 + \mathcal{C}_{\text{beh}}^2, \tag{47}$$

implying a super-additive adversarial burden.

Empirically, semantic counterfactuals over $N = 50$ borderline-malicious samples yield $\mathbb{E}[\mathcal{C}_{\text{sem}}] = 3.27$ with low variance (Table 8), while behavioural counterfactuals exhibit a comparable non-trivial cost. The fusion agent therefore enforces a compounded adversarial requirement: perturbations that are sufficient to evade one modality are generally insufficient to evade the fused decision. This establishes a principled robustness advantage rooted in the geometry of the latent spaces rather than in heuristic thresholding.

Translation of counterfactual costs to real-world executable constraints. The counterfactual norms $\mathcal{C}_{\text{beh}}$ and $\mathcal{C}_{\text{sem}}$ operate in the continuous relaxation of the feature space and must be interpreted in light of the discrete constraints imposed by real malware executables. For the *behavioural agent*, the most perturbed dimensions are LBA and GPA variance statistics (Table 4). Changing these values in a real ransomware binary requires altering the actual I/O access pattern—e.g., replacing bursty sector-sequential writes with randomised access—which directly conflicts with the ransomware's encryption strategy (MITRE ATT&CK T1486: Data Encrypted for Impact). An adversary who reduces behavioural I/O variance sufficiently to cross θ would thereby degrade encryption throughput, creating a practical trade-off between evasion and payload functionality. For the *semantic agent*, the perturbed embedding dimensions correspond to PE structural features—import table structure, section entropy, and resource field layout (Table 7). Altering these requires modifying the binary's PE header, which must remain valid for the OS loader. Standard PE obfuscation tools (e.g., section renaming, import hiding via load-time resolution) can shift embedding coordinates, but they operate under format constraints (valid Magic bytes, consistent SizeOfCode fields) that bound the feasible deformation to a subset of the continuous neighbourhood $\mathcal{N}_\epsilon(\mathbf{z})$. The reported $\mathbb{E}[\mathcal{C}_{\text{sem}}] = 3.27$ therefore represents a lower bound on the real-world adversarial effort; the true cost, under PE validity constraints, is higher.

Failure Modes and Practical Limits

While counterfactual analysis demonstrates strong local robustness, certain failure modes remain theoretically possible. First, the trust-region constrained optimisation in Equation (37) ensures minimal semantic deformation, but does not guarantee semantic validity under extreme distribution shift. An adversary capable of synthesising embeddings that remain statistically plausible while violating real-world PE constraints could, in principle, reduce the effective counterfactual cost.

Second, the observed near-orthogonality between semantic and behavioural counterfactual directions holds locally. In highly non-linear regions of the fusion space, adversarial perturbations that exploit higher-order interactions between modalities may partially circumvent the additive cost structure in Equation (47). Such attacks,

however, require coordinated manipulation of both execution behaviour and binary structure, significantly increasing implementation complexity.

Finally, counterfactual norms quantify minimal effort under continuous relaxation of the input space. Discrete constraints imposed by executable formats, operating-system semantics, and runtime dependencies further restrict feasible adversarial transformations. As a result, the reported counterfactual costs should be interpreted as lower bounds on real-world adversarial effort rather than exact attack recipes.

11. Ablation Studies

This section isolates the contribution of individual architectural components—CTGAN augmentation, uncertainty inputs, and modality contributions—by systematically removing each element and measuring the resulting performance degradation.

The influence of the CTGAN augmentation is quantified by removing it from the fusion stage. If the empirical score distribution is denoted by $\mathcal{D}_{\mathbf{s}}$, CTGAN learns a generator such that

$$\tilde{\mathcal{D}}_{\mathbf{s}} \approx p(\mathbf{s} \mid y), \tag{48}$$

producing synthetic samples that expand the score manifold. Removing CTGAN reduces the RanSMAP AUC from 0.540 to 0.514, indicating that density enrichment of $\mathbf{s}$ contributes to more robust decision boundaries.

Removing uncertainty inputs forces the fusion model to depend only on

$$\mathbf{s}' = (p_{\text{sem}}, p_{\text{beh}}), \tag{49}$$

which eliminates Eq. (3) from the decision mechanism. As a consequence, the number of false negatives increases sharply, since the classifier can no longer separate high-risk but uncertain samples from high-risk confident ones.

Finally, replacing the trained fusion MLP with the naive score mean,

$$\hat{p}_{\text{mean}} = \tfrac{1}{2}\big(p_{\text{sem}} + p_{\text{beh}}\big), \tag{50}$$

performs significantly worse on RanSMAP due to the weak correlation of p_{beh} with ground truth.

A summary of these results is shown in Tables 9 and 10.

Fusion Method	**RDset AUC**	**RanSMAP AUC**
MLP w/ CTGAN	1.0000	0.5397
MLP w/o CTGAN	0.9967	0.5142
Mean Fusion (Eq. 50)	0.884	0.501

Table 9: Ablation of fusion strategies.

Model	**RDset Esc. Rate**	**RanSMAP Esc. Rate**	**FN (RanSMAP)**
Full model	0.72	0.82	43,656
No uncertainty (remove Eq. 3)	0.91	0.97	58,441

Table 10: Effect of removing uncertainty terms on triage performance.

11.1. Ablation Analysis of Modal Contributions

To isolate the contribution of each agent to robustness and explainability, we conduct a conceptual ablation analysis in the fusion space. Let $p_{\text{fus}}(\mathbf{h}_{\text{beh}}, \mathbf{z}_{\text{sem}})$ denote the full fusion prediction. We consider two restricted settings:

$$p_{\text{fus}}^{(-\text{sem})} = p_{\text{fus}}(\mathbf{h}_{\text{beh}}, \mathbf{0}), \tag{51}$$

$$p_{\text{fus}}^{(-\text{beh})} = p_{\text{fus}}(\mathbf{0}, \mathbf{z}_{\text{sem}}). \tag{52}$$

In the absence of semantic information, behavioural counterfactual norms decrease, indicating that certain evasive trajectories become feasible through execution-only manipulations. Conversely, removing behavioural input reduces the counterfactual cost associated with static embedding deformations, enabling purely structural evasion strategies. Neither ablated configuration preserves the super-additive adversarial cost observed in Equation (47).

Importantly, ablation also degrades interpretability consistency. Salient semantic dimensions identified via gradient sensitivity and permutation importance (Equations (34)–(35)) lose explanatory power when behavioural context is removed, and vice versa. This confirms that robustness and interpretability are jointly maximised only when both modalities are present, validating the architectural necessity of the fusion design.

12. Discussion

This section interprets the experimental results in terms of the distributional asymmetry between the two feature spaces, discusses the role of uncertainty in balancing automation and analyst workload, and addresses the computational overhead of the proposed approach.

The experiments highlight a marked asymmetry between the discriminative structures of the semantic and behavioural feature spaces. Let the class-conditioned distributions in semantic space be $p(\mathbf{z} \mid y)$. Their separation can be quantified using the Wasserstein-2 distance

$$d_{\text{sem}} = \mathrm{W}_2(p(\mathbf{z} \mid y=1),\, p(\mathbf{z} \mid y=0))\,, \tag{53}$$

and likewise for behavioural data,

$$d_{\text{beh}} = \mathrm{W}_2(p(\mathbf{b} \mid y=1),\, p(\mathbf{b} \mid y=0))\,. \tag{54}$$

The empirical results strongly suggest that $d_{\text{sem}} \gg d_{\text{beh}}$, explaining why semantic predictions cluster near $\{0, 1\}$ while behavioural predictions exhibit broad dispersion.

The fusion model exploits this asymmetry by treating the semantic component as a strong positive signal whenever uncertainty is low and treating the behavioural component primarily as a modulator of $\sigma_{\max}$. The decision rule therefore becomes highly sensitive to the joint geometry of the fused probability $\hat{p}$ in Eq. (23) and the maximal uncertainty in Eq. (3). Samples with $\hat{p} \geq \tau_{\text{high}}$ and $\sigma_{\max} \leq \kappa_{\text{low}}$ fall into the autonomous containment region, whereas samples with $\hat{p} < \tau$ and $\sigma_{\max} \leq \kappa$ are typically allowed. Intermediate samples reside within the escalation region, which reflects deliberate conservatism in the presence of behavioural ambiguity.

Computational Complexity and Runtime Performance

Agentic SABRE's inference pipeline comprises three sequential stages: (i) agent inference with MC Dropout, (ii) score-level fusion via the Decision Agent MLP, and (iii) threshold evaluation. For agent inference, each of the two CNN agents requires T stochastic forward passes. With $T = 30$ and 1D CNN architectures of depth 3 (kernel size 3, 64 filters), each agent completes a single sample inference in approximately 2–5 ms on a standard CPU, yielding an MC Dropout estimate in $\approx$ 60–150 ms per agent. The Decision Agent MLP (two hidden layers, 64 units each) adds < 1 ms. Threshold evaluation (Eq. 4) is $O(1)$. The total per-sample latency is therefore approximately 0.15–0.35 s on CPU, well within the window of real-time incident triage in SOC environments (typical analyst review time: minutes). On GPU, the T forward passes can be batched, reducing latency by a factor of $\sim T$, bringing per-sample time to < 10 ms. Training times are dominated by CTGAN fitting on the score dataset (typically ~ 200 epochs, $\approx$ 5–10 min on CPU for the score-space dataset sizes used here). The modular architecture means each agent can be retrained independently, and the triage policy can be recalibrated (via threshold optimisation) without any model retraining, making operational updates lightweight.

Overall, the mathematical structure of Agentic SABRE demonstrates that uncertainty-aware evidence fusion is essential for ransomware detection, and that the triage policy cannot be decoupled from the predictive geometry of the agents themselves. The interplay between Eqs. (20)–(3) forms the core mechanism through which the system balances automation, accuracy, and operational safety.

13. Limitations and Future Work

This section acknowledges the principal constraints on the current design—including the conditional independence assumption in the fusion layer, the approximation quality of Monte Carlo Dropout under distribution shift, and the axis-aligned nature of the triage policy—and outlines concrete directions for future research.

Although the results demonstrate the effectiveness of Agentic SABRE, several limitations constrain the system's broader applicability. The fusion mechanism in Eq. (23) assumes that the semantic and behavioural agents generate conditionally independent evidential contributions. In operational settings, semantic metadata and behavioural traces may exhibit dependencies that violate this assumption, potentially altering the joint distribution in Eq. (22). A more expressive fusion function, such as a normalising flow, could model the complete joint likelihood more accurately.

The uncertainty estimates used to compute $\sigma_{\max}$ in Eq. (3) rely on Monte Carlo dropout sampling. While practical, this estimator may under-approximate true epistemic uncertainty in regions where behavioural data exhibit non-stationary distributions. Bayesian neural networks or deep ensembles could provide more calibrated uncertainty but would incur higher computational cost.

The triage policy described in Eqs. (9) and (10) establishes axis-aligned thresholds. Although this provides interpretability and stability, adversarial examples that manipulate the fused score while remaining within tolerance of $\sigma_{\max}$ could produce borderline cases. Future work may instead adopt a decision surface

$$g(\hat{p}, \sigma_{\max}) = 0, \tag{55}$$

where g is a monotone function learned from utility or risk constraints, enabling curved boundaries adapted to the uncertainty structure.

Finally, the behavioural telemetry in RanSMAP reveals that the distributions in Eq. (13) are highly entangled. Future study could investigate hybrid temporal architectures that incorporate attention over micro-bursts of activity to increase separability, thereby widening the Wasserstein gap between malicious and benign trajectories. The modularity of Agentic SABRE allows such components to be integrated without modifying the overarching triage mechanism, making the system suitable for continual architectural refinement. The modular structure of SABRE allows the triage policy to be deployed independently of the underlying detection models, enabling policy-level updates without retraining.

14. Conclusion

This work introduced Agentic SABRE, an uncertainty-aware neuro–symbolic multi-agent architecture for adaptive ransomware detection. The framework integrates semantic and behavioural classifiers, each operating within its own representational manifold, and fuses their outputs through a calibrated decision module enriched by synthetic CTGAN augmentation. The fused probability, defined in Eq. (23), combines two heterogeneous predictive sources, whereas the epistemic uncertainty, articulated through Eq. (3), establishes a principled mechanism for risk-sensitive triage.

The results demonstrate that semantic telemetry yields nearly disjoint class-conditioned distributions, producing fused decision regions with extremely high separability. In contrast, behavioural telemetry exhibits a narrower Wasserstein distance between malicious and benign distributions, as shown by Eqs. (53)–(54), thereby creating high-uncertainty regions that necessitate conservative policies. The system's adaptability is grounded in the triage thresholds of Eq. (25) and the policy optimisation objective of Eq. (12), which jointly determine whether the system should automate containment, escalate for human judgement, or allow execution.

Across two distinct datasets, Agentic SABRE achieves perfect or near-perfect separation where semantic structure dominates, while maintaining robust performance under behavioural variability. The architecture's agentic nature, characterised by self-calibration, evidence fusion, and

structured uncertainty quantification, establishes a computational framework capable not only of detection but also of decision-theoretic reasoning under uncertainty. The incorporation of symbolic cues and the modular design make the system extensible and suitable for continual refinement as ransomware tactics evolve. The results confirm that uncertainty-aware triage is essential for safe deployment of autonomous defence tools in real operational environments.

References

[1] D. W. Fernando, N. Komninos, T. Chen, A study on the evolution of ransomware detection using machine learning and deep learning techniques, IoT 1 (2) (2020) 551–604.

[2] S. Razaulla, C. Fachkha, C. Markarian, A. Gawanmeh, W. Mansoor, B. C. M. Fung, C. Assi, The age of ransomware: A survey on the evolution, taxonomy, and research directions, IEEE Access 11 (2023) 40698–40723. doi:10.1109/ACCESS.2023.3268535.
URL https://doi.org/10.1109/ACCESS.2023.3268535

[3] U. Urooj, B. A. S. Al-Rimy, A. B. Zainal, F. Saeed, A. Abdelmaboud, W. Nagmeldin, Addressing behavioral drift in ransomware early detection through weighted generative adversarial networks, IEEE Access 12 (2024) 3910–3925. doi:10.1109/ACCESS.2023.3348451.
URL https://doi.org/10.1109/ACCESS.2023.3348451

[4] F. Ceschin, M. Botacin, H. M. Gomes, F. A. Pinage, L. S. Oliveira, A. Grégio, Fast & furious: Modelling malware detection as evolving data streams, CoRR abs/2205.12311 (2022). arXiv:2205.12311, doi:10.48550/ARXIV.2205.12311.
URL https://doi.org/10.48550/arXiv.2205.12311

[5] D. W. Fernando, N. Komninos, Fesad ransomware detection framework with machine learning using adaption to concept drift, Comput. Secur. 137 (2024) 103629. doi:10.1016/J.COSE.2023.103629.
URL https://doi.org/10.1016/j.cose.2023.103629

[6] A. Alraizza, A. Algarni, Ransomware detection using machine learning: A survey, Big Data Cogn. Comput. 7 (3) (2023) 143. doi:10.3390/BDCC7030143.
URL https://doi.org/10.3390/bdcc7030143

[7] U. Zahoora, A. Khan, M. Rajarajan, S. H. Khan, M. Asam, T. Jamal, Ransomware detection using deep learning based unsupervised feature extraction and a cost sensitive pareto ensemble classifier, Scientific reports 12 (1) (2022) 15647.

[8] B. G. Doan, D. Q. Nguyen, P. Montague, T. Abraham, O. Y. de Vel, S. Camtepe, S. S. Kanhere, E. Abbasnejad, D. C. Ranasinghe, Bayesian learned models can detect adversarial malware for free, in: J. García-Alfaro, R. Kozik, M. Choras, S. K. Katsikas (Eds.), Computer Security - ESORICS 2024 - 29th European Symposium on Research in Computer Security, Bydgoszcz, Poland, September 16-20, 2024, Proceedings, Part I, Vol. 14982 of Lecture Notes in Computer Science, Springer, 2024, pp. 45–65. doi:10.1007/978-3-031-70879-4_3.
URL https://doi.org/10.1007/978-3-031-70879-4_3

[9] J. Ispahany, M. R. Islam, M. Z. Islam, M. A. Khan, Ransomware detection using machine learning: A review, research limitations and future directions, IEEE Access 12 (2024) 68785–68813. doi:10.1109/ACCESS.2024.3397921.
URL https://doi.org/10.1109/ACCESS.2024.3397921

[10] H. Kabuye, B. Issac, R. Yumlembam, J. Neera, Explainable and uncertainty aware ai-based ransomware detection, IEEE Access 13 (2025) 106573–106589. doi:10.1109/ACCESS.2025.3581424.
URL https://doi.org/10.1109/ACCESS.2025.3581424

[11] A. Alraizza, A. Algarni, A. Alrayzah, Enhancing ransomware detection using storage access pattern, in: I. Maglogiannis, L. S. Iliadis, A. Andreou, A. Papaleonidas (Eds.), Artificial Intelligence Applications and Innovations - 21st IFIP WG 12.5 International Conference, AIAI 2025, Limassol, Cyprus, June 26-29, 2025, Proceedings, Part III, Vol. 757 of IFIP Advances in Information and Communication Technology, Springer, 2025, pp. 57–73. doi:10.1007/978-3-031-96231-8_5.
URL https://doi.org/10.1007/978-3-031-96231-8_5

[12] N. V. Chawla, K. W. Bowyer, L. O. Hall, W. P. Kegelmeyer, SMOTE: synthetic minority over-sampling technique, J. Artif. Intell. Res. 16 (2002) 321–357. doi:10.1613/JAIR.953.
URL https://doi.org/10.1613/jair.953

[13] L. Xu, M. Skoularidou, A. Cuesta-Infante, K. Veeramachaneni, Modeling tabular data using conditional GAN, in: H. M. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. B. Fox, R. Garnett (Eds.), Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada, 2019, pp. 7333–7343.
URL https://proceedings.neurips.cc/paper/2019/hash/254ed7d2de3b23ab10936522dd547b78-Abstract.html

[14] Y. Gal, Z. Ghahramani, Dropout as a bayesian approximation: Representing model uncertainty in deep learning, in: M. Balcan, K. Q. Weinberger (Eds.), Proceedings of the 33nd International Conference on Machine Learning, ICML 2016, New York City, NY, USA, June 19-24, 2016, Vol. 48 of JMLR Workshop and Conference Proceedings, JMLR.org, 2016, pp. 1050–1059.
URL http://proceedings.mlr.press/v48/gal16.html

[15] J. Gawlikowski, C. R. N. Tassi, M. Ali, J. Lee, M. Humt, J. Feng, A. M. Kruspe, R. Triebel, P. Jung, R. Roscher, M. Shahzad, W. Yang, R. Bamler, X. Zhu, A survey of uncertainty in deep neural networks, Artif. Intell. Rev. 56 (S1) (2023) 1513–1589. doi:10.1007/S10462-023-10562-9.
URL https://doi.org/10.1007/s10462-023-10562-9

[16] T. R. Besold, A. S. d'Avila Garcez, S. Bader, H. Bowman, P. M. Domingos, P. Hitzler, K. Kühnberger, L. C. Lamb, P. M. V. Lima, L. de Penning, G. Pinkas, H. Poon, G. Zaverucha, Neural-symbolic learning and reasoning: A survey and interpretation, in: P. Hitzler, M. K. Sarker (Eds.), Neuro-Symbolic Artificial Intelligence: The State of the Art, Vol. 342 of Frontiers in Artificial Intelligence and Applications, IOS Press, 2021, pp. 1–51. doi:10.3233/FAIA210348.
URL https://doi.org/10.3233/FAIA210348

[17] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, in: I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, R. Garnett (Eds.), Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA, 2017, pp. 5998–6008.
URL https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

[18] S. Tian, T. Zhang, J. Liu, J. Wang, X. Wu, X. Zhu, R. Zhang, W. Zhang, Z. Yuan, S. Mao, D. I. Kim, Exploring the role of large language models in cybersecurity: A systematic survey, IEEE Transactions on Network Science and EngineeringEarly Access (2025).
URL https://arxiv.org/abs/2504.15622

[19] J. Zhang, P. Wu, J. London, D. Tenney, Benchmarking and evaluating large language models in phishing detection for small and midsize enterprises: A comprehensive analysis, IEEE Access 13 (2025) 28335–28352. doi:10.1109/ACCESS.2025.3540075.
URL https://doi.org/10.1109/ACCESS.2025.3540075

[20] U. Patel, F. Yeh, C. Gondhalekar, CANAL - cyber activity news alerting language model : Empirical approach vs. expensive llms, in: 3rd IEEE International Conference on AI in Cybersecurity, ICAIC 2024, Houston, TX, USA, February 7-9, 2024, IEEE, 2024, pp. 1–12. doi:10.1109/ICAIC60265.2024.10433839.
URL https://doi.org/10.1109/ICAIC60265.2024.10433839

[21] I. Ashrapov, Tabular gans for uneven distribution, CoRR abs/2010.00638 (2020). arXiv:2010.00638.
URL https://arxiv.org/abs/2010.00638

[22] O. Habibi, M. Chemmakha, M. Lazaar, Imbalanced tabular data modelization using CTGAN and machine learning to im-

prove iot botnet attacks detection, Eng. Appl. Artif. Intell. 118 (2023) 105669. doi:10.1016/J.ENGAPPAI.2022.105669.
URL https://doi.org/10.1016/j.engappai.2022.105669

[23] B. N. Chaithanya, S. H. Brahmananda, Ai-enhanced defense against ransomware within the organization's architecture, J. Cyber Secur. Mobil. 11 (4) (2022) 621–654. doi:10.13052/JCSM2245-1439.1146.
URL https://doi.org/10.13052/jcsm2245-1439.1146

[24] Z. Zhang, S. Li, L. Zhang, J. Ye, C. Hu, L. Yan, LLM-LADE: large language model-based log anomaly detection with explanation, Knowl. Based Syst. 326 (2025) 114064. doi:10.1016/J.KNOSYS.2025.114064.
URL https://doi.org/10.1016/j.knosys.2025.114064

[25] S. M. Lundberg, S. Lee, A unified approach to interpreting model predictions, in: I. Guyon, U. von Luxburg, S. Bengio, H. M. Wallach, R. Fergus, S. V. N. Vishwanathan, R. Garnett (Eds.), Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA, 2017, pp. 4765–4774.
URL https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[26] S. Jia, Z. Cui, K. Wang, Y. Ding, D. Han, T. Ma, Adversarial and cooperation tasks in multi-agent system with large language models, in: 2024 IEEE International Conference on Unmanned Systems (ICUS), IEEE, 2024, pp. 1840–1847.

[27] M. Soltani, K. Khajavi, M. J. Siavoshani, A. H. Jahangir, A multi-agent adaptive deep learning framework for online intrusion detection, CoRR abs/2303.02622 (2023). arXiv:2303.02622, doi:10.48550/ARXIV.2303.02622.
URL https://doi.org/10.48550/arXiv.2303.02622

[28] W. Yang, L. Some, M. Bain, B. H. Kang, A comprehensive survey on integrating large language models with knowledge-based methods, Knowl. Based Syst. 318 (2025) 113503. doi:10.1016/J.KNOSYS.2025.113503.
URL https://doi.org/10.1016/j.knosys.2025.113503

[29] I. A. Fares, M. A. Elaziz, Explainable tabnet transformer-based on google vizier optimizer for anomaly intrusion detection system, Knowl. Based Syst. 316 (2025) 113351. doi:10.1016/J.KNOSYS.2025.113351.
URL https://doi.org/10.1016/j.knosys.2025.113351

[30] M. Hirano, R. Hodota, R. Kobayashi, Ransap: An open dataset of ransomware storage access patterns for training machine learning models, Digit. Investig. 40 (Supplement) (2022) 301314. doi:10.1016/J.FSIDI.2021.301314.
URL https://doi.org/10.1016/j.fsidi.2021.301314

[31] M. Hirano, R. Kobayashi, Ransmap: Open dataset of ransomware storage and memory access patterns for creating deep learning based ransomware detectors, Comput. Secur. 150 (2025) 104202. doi:10.1016/J.COSE.2024.104202.
URL https://doi.org/10.1016/j.cose.2024.104202

[32] M. Mathur, Ransomware (malware) detection using machine learning, accessed: Nov. 9, 2023 (2020).
URL https://github.com/muditmathur2020/RansomwareDetection/tree/master

Algorithm 1 Agentic SABRE: Training and Deployment Pipeline

Input (training): Raw semantic data D_{sem}, behavioural logs D_{beh}
Input (deployment): New sample $x_{\text{sem}}, x_{\text{beh}}$
Output (deployment): Decision $\in \{\text{AUTO-CONTAIN}, \text{ESCALATE}, \text{ALLOW}\}$

// Phase 1: Hybrid feature extraction
Extract semantic embeddings $\mathbf{z}_i$ from D_{sem} using encoder $\mathcal{E}(\cdot)$
Extract behavioural feature vectors $\mathbf{b}_i$ from D_{beh} via sliding-window statistics

// Phase 2: Feature-level augmentation
Apply SMOTE to $\{\mathbf{z}_i\}$ and $\{\mathbf{b}_i\}$ independently to address class imbalance

// Phase 3: Train base agents with MC Dropout
Train Semantic Agent CNN A_{sem} on $\{\mathbf{z}_i\}$
Train Behavioural Agent CNN A_{beh} on $\{\mathbf{b}_i\}$

// Phase 4: Score-level augmentation and Decision Agent
for each training sample i **do**
    Compute mean scores $\bar{p}_{\text{sem}}^{(i)}$, $\bar{p}_{\text{beh}}^{(i)}$ via MC Dropout (Alg. 2)
    Form score vector $s_i = [\bar{p}_{\text{sem}}^{(i)}, \bar{p}_{\text{beh}}^{(i)}]$
end for
Train CTGAN on $\{(s_i, y_i)\}$ and obtain augmented scores $\tilde{\mathcal{D}}_S$ (Alg. 3)
Train Decision Agent A_{dec} on $\mathcal{D}_S \cup \tilde{\mathcal{D}}_S$

// Phase 5: Deployment-time inference and triage
Obtain $(\bar{p}_{\text{sem}}, \sigma_{\text{sem}})$ for x_{sem} via MC Dropout
Obtain $(\bar{p}_{\text{beh}}, \sigma_{\text{beh}})$ for x_{beh} via MC Dropout
Form $s = [\bar{p}_{\text{sem}}, \bar{p}_{\text{beh}}]$ and compute fused risk $\hat{p} = A_{\text{dec}}(s)$
Compute aggregated uncertainty $\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}})$
Apply triage policy using thresholds $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$ (Alg. 4)

Algorithm 2 MC Dropout Uncertainty Estimation for a Neural Agent

Input: Trained neural agent A with dropout layers, sample x, number of passes T
Output: Predictive mean $\bar{p}$, epistemic uncertainty σ
Set A to evaluation mode
Activate dropout layers for inference (e.g., set dropout modules to `train()` only)
for $t = 1$ to T **do**
    $p_t \leftarrow A(x)$ ▷ stochastic forward pass with dropout
end for
Compute predictive mean:

$$\bar{p} = \frac{1}{T} \sum_{t=1}^{T} p_t$$

Compute predictive standard deviation:

$$\sigma = \sqrt{\frac{1}{T} \sum_{t=1}^{T} (p_t - \bar{p})^2}$$

return $(\bar{p}, \sigma)$

Algorithm 3 Score-Level CTGAN Augmentation for Decision Agent Training

Input: Trained base agents $A_{\text{sem}}, A_{\text{beh}}$; training features $\{\mathbf{z}_i\}, \{\mathbf{b}_i\}$; labels $\{y_i\}$
Output: Augmented score dataset $\tilde{\mathcal{D}}_S$, for training A_{dec}
// Step 1: Collect score tuples
for each training sample i **do**
    Compute $(\bar{p}^{(i)}_{\text{sem}}, \sigma^{(i)}_{\text{sem}})$ via Alg. 2 on $\mathbf{z}_i$
    Compute $(\bar{p}^{(i)}_{\text{beh}}, \sigma^{(i)}_{\text{beh}})$ via Alg. 2 on $\mathbf{b}_i$
    Form score vector $s_i = [\bar{p}^{(i)}_{\text{sem}}, \bar{p}^{(i)}_{\text{beh}}]$
end for
Construct score dataset $\mathcal{D}_S = \{(s_i, y_i)\}$
// Step 2: Train CTGAN in score space
Train CTGAN generator G and discriminator D on $\mathcal{D}_S$ conditioned on labels y_i
Generate synthetic score samples $\tilde{s}_j = G(z_j, y_j)$ with noise z_j and labels y_j
Form augmented set $\tilde{\mathcal{D}}_S = \{(\tilde{s}_j, y_j)\}$
// Step 3: Train Decision Agent
Train Decision Agent A_{dec} on $\mathcal{D}_S \cup \tilde{\mathcal{D}}_S$
return $\tilde{\mathcal{D}}_S$

Algorithm 4 Uncertainty-Aware Triage Policy

Input: Fused risk $\hat{p}$, agent uncertainties $\sigma_{\text{sem}}, \sigma_{\text{beh}}$
Input: Risk thresholds τ, τ_{high}; uncertainty thresholds κ, κ_{low}
Output: Decision $\in \{\text{AUTO-CONTAIN}, \text{ESCALATE}, \text{ALLOW}\}$
Compute aggregated uncertainty $\sigma_{\max} \leftarrow \max(\sigma_{\text{sem}}, \sigma_{\text{beh}})$
if $\hat{p} \geq \tau_{\text{high}}$ **and** $\sigma_{\max} \leq \kappa_{\text{low}}$ **then**
    return AUTO-CONTAIN ▷ high risk, low uncertainty
else if $\hat{p} \geq \tau$ **or** $\sigma_{\max} > \kappa$ **then**
    return ESCALATE ▷ risky or uncertain
else
    return ALLOW ▷ benign and confident
end if